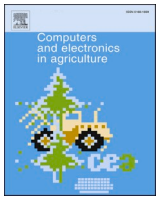
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Review Article

Research progress of autonomous navigation path planning methods for agricultural ground machinery: a review

Xuepeng Xiao [1], Hongbo Yuan [1], Yashuai Yang, Man Cheng [*]

College of Mechanical and Electrical Engineering, Agricultural University of Hebei, Baoding 071000, China

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ABSTRACT

Autonomous navigation of agricultural ground machinery is a key enabler of large-scale, precise, and sustainable crop production. In the past few years, numerous path-planning solutions have been proposed for tractors and unmanned ground vehicles (UGVs) operating in open fields, orchards, and protected environments. However, the evidence remains fragmented across scenarios and information sources, and a scenario-aware synthesis is still limited. This review systematically maps and analyses research on autonomous navigation and path planning for agricultural ground machinery published between January 2019 and September 2025. Following Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 and the Preferred Reporting Items for Systematic Reviews and Meta-Analyses extension for Scoping Reviews (PRISMA-ScR), 564 records were screened and 210 studies were included. Studies were coded by the dominant planning information (static, real-time, or hybrid), operation mode (single-vehicle or multi-vehicle), scenario, platform, sensing configuration, and validation strategy. Descriptive statistics, cross-tabulation, and keyword co-occurrence analysis were used to synthesize the landscape. Results indicate that single-vehicle systems predominantly rely on real-time information, particularly in open-field and orchard operations, whereas multi-vehicle studies still mainly adopt static, map-based coverage planning. Hybrid approaches that integrate global maps with online perception, navigation, and control are emerging, with increasing attention to complex environments such as orchards and greenhouses. Across scenarios, the dominant technical trajectory evolves from static global coverage planning to real-time local planning and obstacle avoidance, and further to hybrid, multi-source fusion strategies. This review provides a scenario-aware framework that summarises representative algorithms and sensing–control configurations, benchmarks them against commercial systems, and identifies key challenges in perception robustness, information fusion, cooperative planning, evaluation protocols, and deployment feasibility. The resulting insights support the design of practical autonomous navigation stacks and help prioritise future research on path planning for agricultural ground machinery.

1. Introduction

Agricultural production faces increasing pressure to improve productivity and product quality while reducing inputs, environmental impacts, and labour dependence (Pak et al., 2022). Achieving this transition at scale requires autonomous ground machinery that can navigate safely and efficiently in real-world farms rather than in controlled test sites. Compared with structured roads or factory settings, agricultural environments introduce persistent operational uncertainty, including uneven and deformable terrain, seasonally evolving crop geometry, degraded Global Navigation Satellite System (GNSS) reception under crop canopies or near buildings, and frequent interactions with humans, animals, and other machines. Under these conditions, robust navigation and path planning become core prerequisites for autonomy.

For agricultural ground machinery, navigation rarely reduces to a single algorithmic module. Instead, it spans an end-to-end perception–planning–control pipeline that must be tailored to specific tasks and scenarios. Representative tasks include complete coverage of irregular arable fields for sowing or spraying (Chakraborty et al., 2022), safe guidance along tree rows in orchards and vineyards, precise manoeuvring and docking within narrow greenhouse aisles, and long-distance transport along farm roads with intermittent field entries.

* Corresponding author.
E-mail address: chengman1982@163.com (M. Cheng).
[1] These authors contributed equally to this work.

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0168-1699/© 2026 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

These scenarios impose distinct constraints on path smoothness, coverage efficiency, turning behaviour, obstacle avoidance, localisation accuracy, and interaction with implements. In addition, farms are increasingly coordinating multiple machines and implements, shifting path planning from single-vehicle guidance to fleet-level task allocation, scheduling, and motion coordination (Mammarella et al., 2022).

Over the past decade, research has advanced (i) global coverage planning for irregular fields, (ii) perception-driven navigation for rows and obstacles, (iii) localisation and simultaneous localisation and mapping (SLAM) with multi-sensor fusion for GNSS-degraded orchards and greenhouses, and (iv) motion control for diverse vehicle kinematics. Existing reviews have provided valuable module-centric summaries of agricultural robotics and navigation (e.g., sensing, mapping, planning, and control). However, for agricultural ground machinery, it remains unclear how planning solutions systematically vary with operational scenarios and, critically, the type of environmental information available to the planner.

From the perspective of path planning for agricultural ground machinery, prior reviews reveal four practical limitations. First, the literature is commonly organised by technical modules or robot categories, with limited emphasis on how planning strategies change across scenarios such as open fields, orchards, greenhouses, and mixed farm environments. Second, many classifications focus on algorithm families without explicitly distinguishing the information that drives planning—static maps and pre-defined waypoints, real-time perception of rows/obstacles/terrain, or hybrid schemes that combine both (Wei, 2021). Third, single-vehicle navigation and multi-vehicle cooperative planning are often treated together, despite their fundamentally different architectures, constraints, and validation requirements. Fourth, several reviews predate recent advances in multi-sensor fusion, SLAM-based perception, and coordinated operation of multiple ground machines, and therefore do not fully capture how these trends are reshaping deployable navigation stacks. As a result, it remains difficult for researchers and practitioners to determine which planning approaches best match specific agricultural scenarios, information sources, and operation modes, or where the most consequential research gaps lie.

To address these issues, this article presents a scoping review of autonomous navigation and path planning methods for agricultural ground machinery published between 2019 and 2025. The review Following the PRISMA 2020 statement (Page et al., 2021) and the PRISMA-ScR extension (Tricco et al., 2018) and focuses on ground platforms in agricultural contexts, including tractors, self-propelled sprayers, carrier robots, and specialised harvesters. A total of 210 studies are included. Rather than conducting a formal *meta*-analysis, the review systematically maps how environmental information types and operation modes are adopted in recent navigation systems, how these choices vary across scenarios, and how research prototypes relate to commercial solutions already deployed in the field.

This review makes three main contributions by extracting cross-scenario patterns and trends from the 2019 to the 2025 literature. First, we propose a two-dimensional analytical framework (information source × operation mode) and quantify the resulting landscape using descriptive statistics and cross-tabulations. Second, we synthesise single-vehicle navigation as an evolution from static, map-based global coverage planning to perception-driven local planning and, increasingly, SLAM- and multi-sensor-fusion-enabled planning and control. Third, we consolidate multi-vehicle cooperative navigation under the same framework, distinguish static coordination from real-time and hybrid cooperation, and benchmark research prototypes against commercial systems to identify mature technologies and system-level bottlenecks that motivate future research.

A central feature of this work is the two-dimensional framework that organises the literature by (i) the dominant information source used for path planning—static, real-time, or hybrid—and (ii) the operation mode—single-vehicle or multi-vehicle. Static-information methods primarily rely on pre-acquired maps, field boundaries, or waypoints; real-time methods base planning mainly on online perception; and hybrid methods combine both sources. Single-vehicle systems target the autonomy of individual machines, whereas multi-vehicle systems coordinate fleets of machines or implements in space and time. Crossing these two dimensions yields six conceptual categories (static-single, static-multi, real-time-single, real-time-multi, hybrid-single, and hybrid-multi) that reflect common engineering decisions in agricultural deployments. The framework underpins the quantitative synthesis, the scenario-based technical review, and the identification of research gaps.

Accordingly, the review addresses three questions: (1) How are the static, real-time, and hybrid information sources used for path planning across open fields, orchards, greenhouses, and mixed farm scenarios? (2) How do information types interact with single- and multi-vehicle operation modes to shape the emerging patterns in sensing configurations, algorithm families, and validation practices? (3) Where are the key gaps and opportunities, particularly for hybrid-information and multi-vehicle systems in complex agricultural environments? Rather than enumerating algorithms, the review uses the framework to extract cross-scenario patterns and to clarify where recent advances translate—or fail to translate—into deployable systems.

The remainder of this article is organised as follows. Section 2 details the PRISMA-ScR methodology, including the search strategy, inclusion criteria, and coding procedures. Section 3 introduces the main scenarios and platforms, defines information-source and operation-mode categories, and summarises the quantitative landscape together with representative commercial systems. Sections 4 and 5 review single-vehicle and multi-vehicle navigation, respectively, emphasising the technical trajectory from static planning to perception-driven and fusion-enabled autonomy. Section 6 provides a cross-cutting discussion of information fusion, environmental diversity, machinery characteristics, farm-scale differences, and system integration, and outlines key bottlenecks and future directions. Section 7 concludes the review.

2. Methods

This review followed PRISMA 2020 and used PRISMA-ScR as guidance for scoping reviews. The objective was not to conduct a formal *meta*-analysis, but to systematically map recent research on autonomous navigation and path planning for agricultural ground machinery, identify recurring patterns, and examine trends across methods, information sources, and operation modes. A PRISMA-ScR flow diagram is provided to document study identification, screening, eligibility assessment, and inclusion, thereby ensuring transparency. The review workflow comprised: (i) defining the research questions and scope; (ii) specifying inclusion and exclusion criteria; (iii) conducting database searches and backward/forward snowballing; (iv) screening records; and (v) extracting and coding study attributes for descriptive statistics and structured narrative synthesis.

2.1. Data sources

The Web of Science Core Collection (Science Citation Index Expanded, SCI-Expanded) served as the primary data source, as it covers major journals in engineering, robotics, and agricultural sciences. To reduce the risk of omitting relevant studies, the database search was complemented by backward and forward snowballing. Backward snowballing screened the reference lists of included studies, whereas forward snowballing identified subsequent publications that cited the included studies.

2.2. Data search strategy

We focused on studies published between January 2019 and September 2025 to capture recent developments in agricultural robotics and autonomous navigation while maintaining relevance to current technologies. Searches targeted titles, abstracts, and author keywords.

The search strategy for autonomous navigation and path planning in agricultural ground machinery was based on Boolean queries. A representative Web of Science query (Topic, TS) was: TS = (agriculture* NEAR/3 (robot* OR vehicle* OR machinery* OR platform)) AND ("path planning" OR navigation OR guidance) AND (field OR orchard OR greenhouse OR farm*). Queries were iteratively refined by expanding synonyms (e.g., adding "guidance system" alongside "navigation") and, where appropriate, controlled vocabulary terms. Search results were exported, de-duplicated, and screened. Backward and forward snowballing were then applied to identify additional relevant studies. Manual screening excluded irrelevant records, such as studies on unmanned aerial vehicles (UAVs), aquatic robots, or theoretical algorithms without agricultural applications.

2.3. Inclusion and exclusion criteria

Studies were included if they met all of the following criteria: (i) peer-reviewed journal articles written in English and published between 2019 and 2025; (ii) focused on agricultural ground platforms or machinery (e.g., tractors, UGVs, or carrier platforms) and addressed at least one component of autonomous navigation or path planning (e.g., localisation, perception, path generation, tracking control, or system integration); (iii) reported quantitative performance indicators or clearly described evaluation outcomes (e.g., localisation error, coverage efficiency, travel distance, energy consumption, or safety-related measures); and (iv) provided sufficient methodological detail and validation in simulation, laboratory, greenhouse, or field settings.

Studies were excluded if they: (i) were not related to agricultural contexts or to path planning for agricultural ground machinery; (ii) were conceptual studies without implementable methods or validation; (iii) were non-peer-reviewed sources; or (iv) were duplicates or earlier versions of the same study.

2.4. Study selection and screening (PRISMA-ScR)

All retrieved records underwent three-stage screening: title screening, abstract screening, and full-text assessment. At each stage, the eligibility criteria in Section 2.3 were applied. Irrelevant records were removed, and reasons for exclusion (e.g., non-agricultural domain, no ground machinery, or no navigation component) were recorded.

After de-duplication and screening, 210 studies on autonomous navigation and path planning for agricultural ground machinery were included. The PRISMA-ScR flow diagram (Fig. 1) summarises the record counts at each stage. Observations on reporting completeness (e.g., clarity of platform description, and availability of performance indicators) were documented but did not affect inclusion if eligibility criteria were met.

2.5. Data extraction and coding

A structured data-extraction template was designed to support consistent comparison across studies. Each study was coded using predefined variables, including: (i) context: crop type, operational scenario (open field, orchard, greenhouse, or other), geographical region (e.g., Asia, Europe, or North America), and platform type (e.g., tractor-mounted implement, self-propelled UGV, or carrier platform); (ii) sensing and computing: primary sensors (e.g., real-time kinematic GNSS (RTK-GNSS), inertial measurement unit (IMU), light detection and ranging (LiDAR), or cameras), sensor-fusion approach, and computing hardware (e.g., embedded controller, industrial PC (IPC), or NVIDIA Jetson); (iii) technical modules: localisation/SLAM, path planning algorithms (e.g., coverage-based methods, graph-search methods, such as

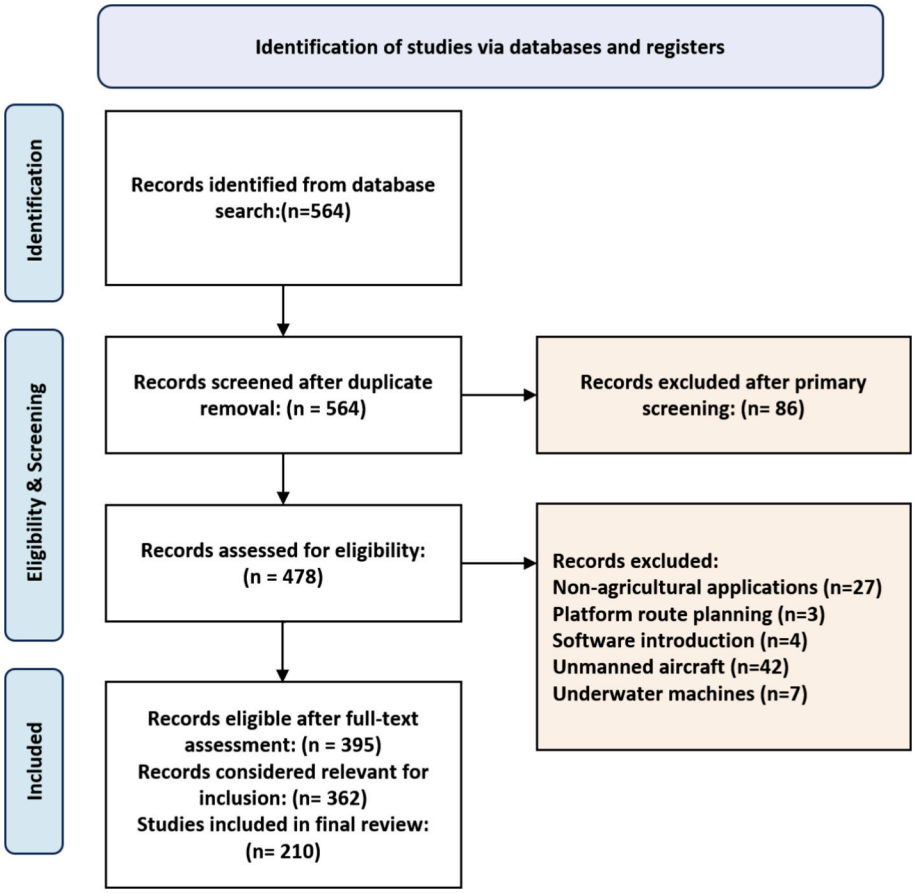


Fig. 1. PRISMA-ScR flow diagram of study identification, screening, eligibility assessment, and inclusion.

A* search, sampling-based methods such as rapidly exploring random tree (RRT), optimisation-based methods, or learning-based methods), obstacle-avoidance strategy, and path-tracking controller (e.g., Pure Pursuit, model predictive control (MPC), sliding-mode control (SMC), or fuzzy control (FC)); and (iv) validation and outcomes: validation setting (simulation, laboratory, greenhouse, or field), key performance indicators (KPIs) (e.g., localisation error, coverage efficiency, or path length, execution time), and reported evidence on robustness and safety.

Each study was coded by one author and cross-checked by another to ensure consistency. Discrepancies were resolved through discussion. Studies were also labelled as primary research articles or review papers to support analyses that distinguish evidence generation from evidence synthesis. The resulting dataset underpins the descriptive statistics in Section 3.4 and the comparative synthesis in Sections 4–6.

2.6. Analytical framework

To synthesise diverse methods systematically, we developed a two-dimensional taxonomy based on: (i) the dominant environmental information source used for planning and (ii) the operation mode of the machinery. Regarding information source, studies were classified as static-information-based (planning primarily driven by pre-acquired maps/boundaries/waypoints), real-time-information-based (planning primarily driven by online perception), or hybrid-information-based (explicit integration of prior information and online perception). Regarding operation mode, the framework distinguishes single-vehicle systems (autonomy of an individual machine) from multi-vehicle systems (explicit coordination of multiple machines and/or implements in space and time).

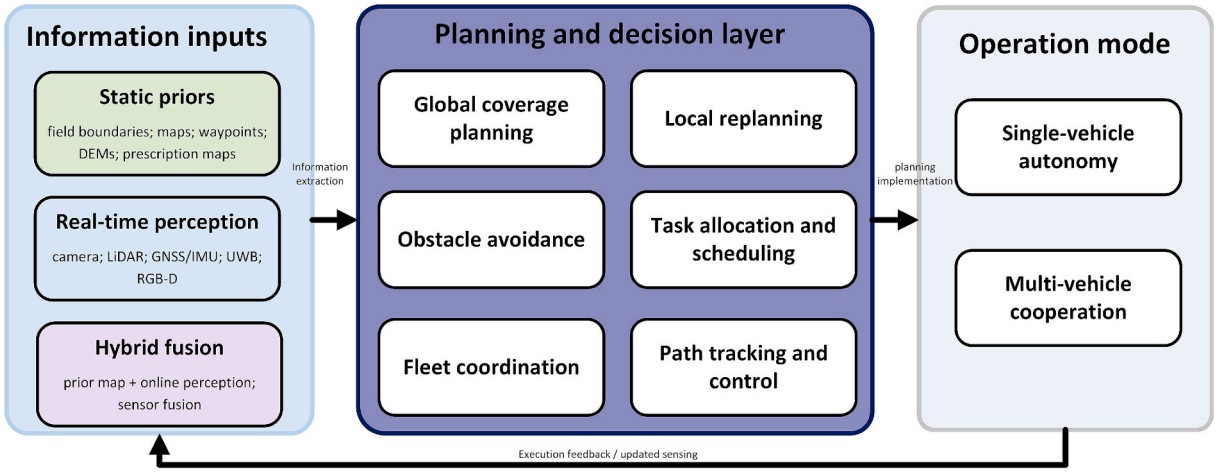
Crossing these dimensions yields six conceptual categories—static-single, static-multi, real-time-single, real-time-multi, hybrid-single, and hybrid-multi—as illustrated in Fig. 2. This framework reflects common engineering decisions in agricultural deployments and provides the basis for the quantitative landscape (Section 3.4) and the technical syntheses (Sections 4 and 5).

To improve reproducibility, coding followed explicit decision rules. The information source was determined by the dominant information that drives path generation or online path updates, rather than by the mere presence of sensors. Studies were coded as static when the operational path was generated offline from prior information (e.g., field boundary maps, waypoints, digital elevation models (DEM), or prescription/task maps) and executed online via tracking control, even if GNSS/IMU provided real-time localisation. Studies were coded as real-time when online sensing (e.g., vision, LiDAR, radar, or ultrasonic) directly supported obstacle detection, crop-row/road perception, and online path generation or replanning. Studies were coded as hybrid when an offline global plan (and/or task allocation) was combined with perception-driven local adjustment, obstacle avoidance, or replanning. For SLAM-based systems, studies were coded as real-time when maps were built online and directly used for local planning, and as hybrid when SLAM was integrated with explicit prior maps or global guidance. Operation mode was coded as multi-vehicle only when explicit coordination was included (e.g., task allocation, conflict resolution, formation control, shared mapping, or communication); otherwise, it was coded as single-vehicle.

2.7. Evidence synthesis and descriptive statistical analysis

The included studies were heterogeneous in sensing configurations, platforms, terrain conditions, planning algorithms, and KPI definitions. Under such heterogeneity, conventional *meta*-analysis or *meta*-regression would be inappropriate and may yield misleading quantitative summaries. We therefore adopted a descriptive, narrative synthesis strategy consistent with scoping-review objectives.

First, descriptive statistics summarised the coded variables. Based on the framework in Section 2.6, we quantified the number of studies per year by information source, the distribution of operation modes across scenarios, and the study counts in each of the six information-source–operation-mode categories. These statistics were computed for all 210 included studies. For the cross-tabulation in Table 1, we retained 185 primary research articles that explicitly reported both a dominant information source and an operation mode, excluding review papers and



Two-dimensional analytical categories

	Single-vehicle autonomy	Multi-vehicle cooperation
Static information	Static-single	Static-multi
Real-time information	Real-time-single	Real-time-multi
Hybrid information	Hybrid-single	Hybrid-multi

Fig. 2. Data-flow-oriented two-dimensional analytical framework for agricultural ground-machinery navigation and path planning.

Table 1
Aggregated cross-tabulation of information types and operation modes based on the six-category analytical framework (primary studies, n = 185).

Information type	Multi-vehicle	Single-vehicle
Hybrid	6	13
Real-time	3	117
Static	12	34

simulation-only studies. Figures and tables in Section 3.4 visualise temporal trends, scenario-dependent distributions, representative KPI patterns, algorithm–scenario relationships, and sensor–scenario configurations across the reviewed studies. To keep the main text focused on cross-study synthesis rather than study-by-study enumeration, the detailed coding records were organised in Appendix Tables A1-A9.

Second, within each analytical category (e.g., static single-vehicle coverage planning, real-time local planning, hybrid global–local planning, or multi-vehicle coordination), we conducted a structured narrative synthesis. Studies were grouped by scenario, sensing configuration, algorithm class, and validation setting, and their technical choices and reported performance were compared qualitatively. Rather than aggregating effect sizes, we emphasised recurring design patterns (e.g., typical sensor combinations in orchards versus open fields; trade-offs between coverage completeness and turning cost) and representative case studies.

Third, as a supplementary perspective on research topics, we conducted a keyword co-occurrence analysis using VOSviewer (v1.6.20). The co-occurrence network and density map (Figs. 3 and 4) were used to identify thematic clusters and to cross-check whether the major technical pathways identified via coding and narrative synthesis (e.g., perception–planning–control pipelines, and multi-source information fusion) were consistent with the keyword landscape. This bibliometric analysis served for orientation only and did not affect study inclusion or quantitative conclusions.

3. Navigation problems, application scenarios and analytical framework

Recent research on autonomous navigation for agricultural ground machinery spans diverse crops, field conditions, and production scales. Three elements jointly shape agricultural navigation: (i) the operational environment (scenario), (ii) the production task (operation objective), and (iii) the information available to the navigation stack (information type and fusion), together with the machinery operation mode. Building

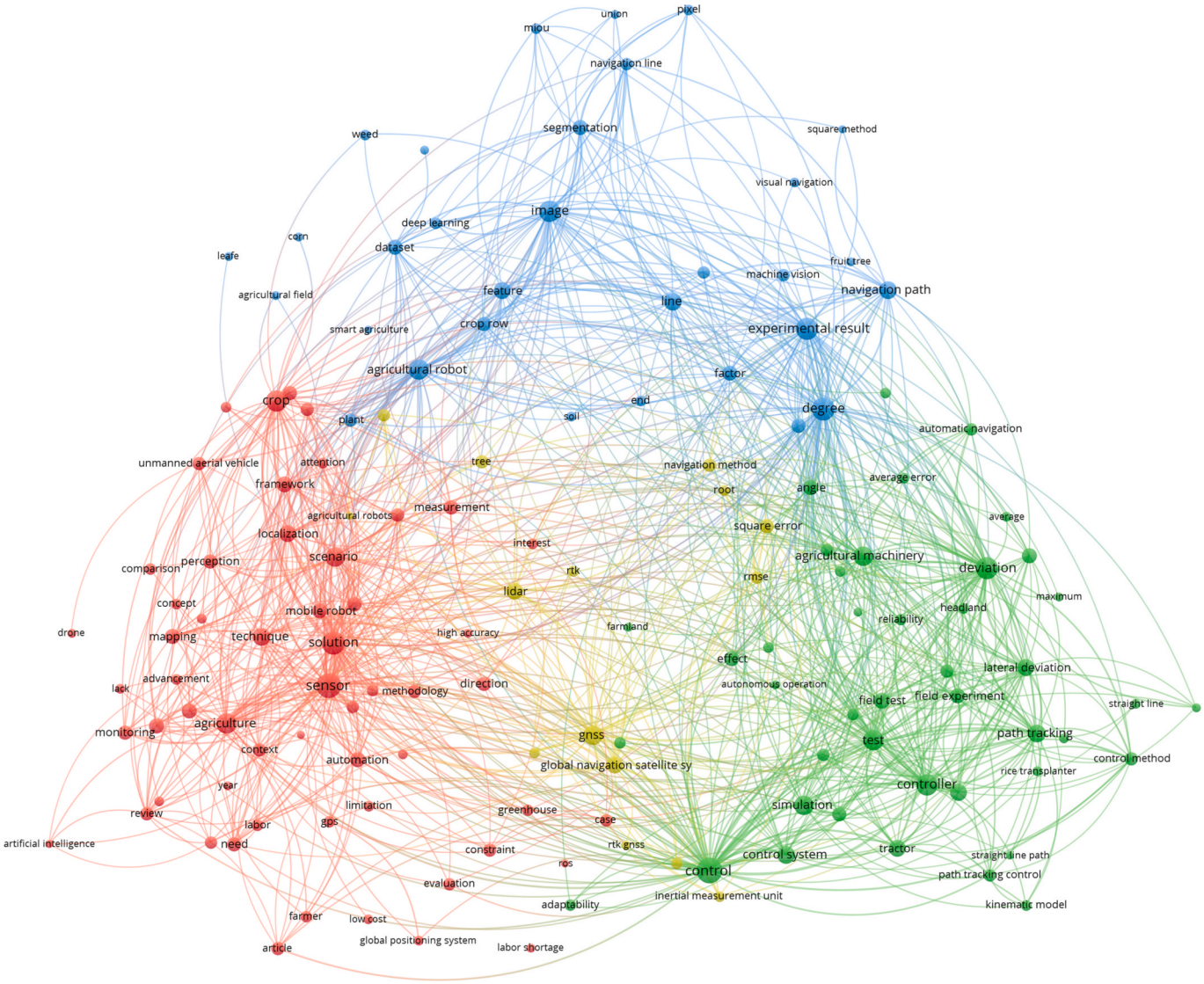


Fig. 3. Keyword co-occurrence network (VOSviewer), where node size, edge width, and colour represent keyword frequency, co-occurrence strength, and semantic clusters, respectively.

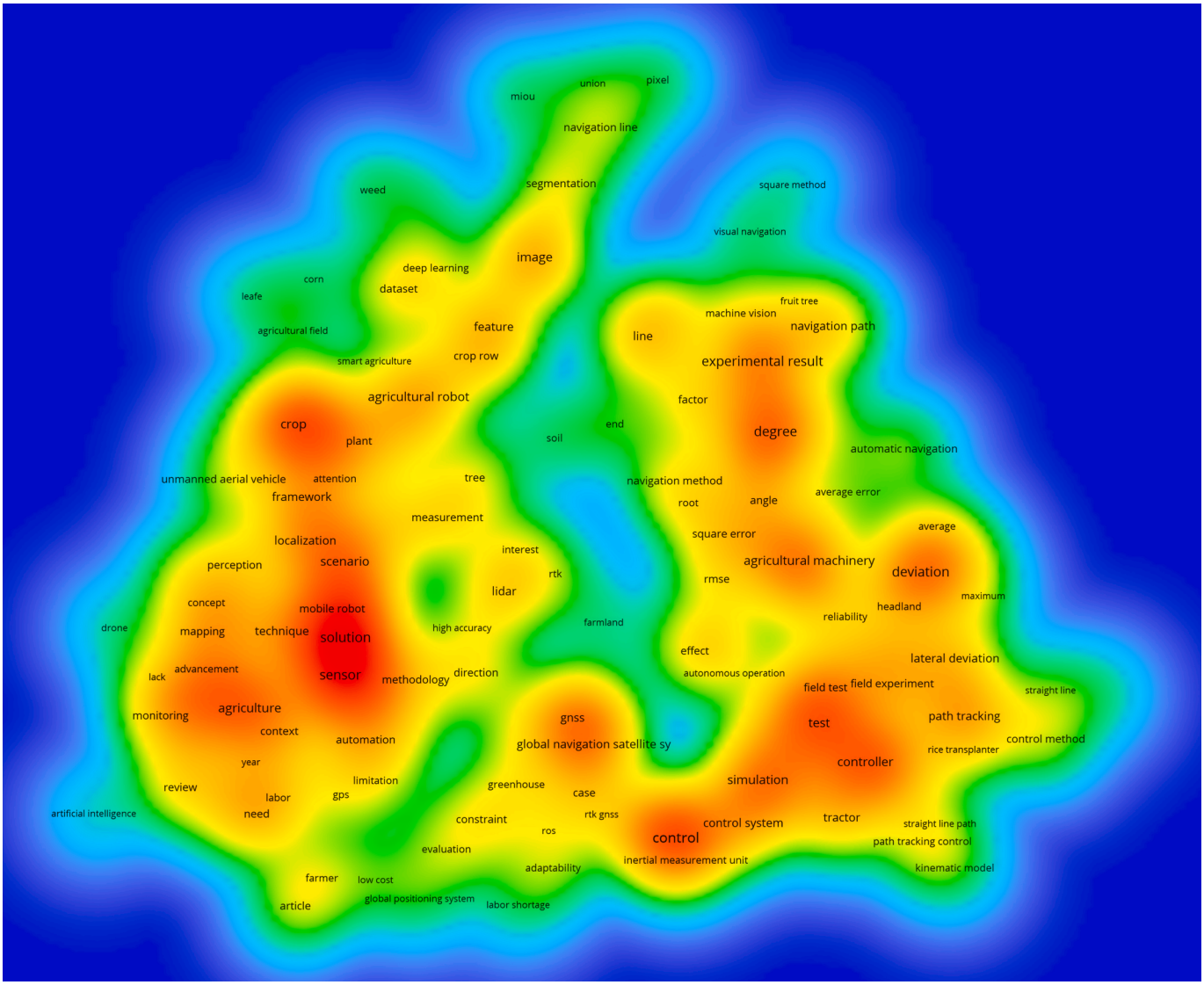


Fig. 4. Keyword density map (VOSviewer), with colour gradient indicating local frequency and co-occurrence density (red = higher, blue = lower).

on the screened studies and the coding scheme in Section 2, this section: (i) summarises major operational scenarios and navigation tasks; (ii) outlines typical machinery platforms; (iii) introduces the two-dimensional framework that organises the remainder of the review; (iv) reports the quantitative landscape of the included studies; and (v) links the findings to commercial navigation solutions.

3.1. Operational scenarios and navigation tasks

The reviewed literature covers four primary operational scenarios: open-field arable farming; orchards and vineyards; protected cultivation (greenhouses and plant factories); and mixed or facility-based environments (e.g., farm roads, livestock barns, or logistics yards). Within each scenario, navigation problems are defined by task-specific constraints and performance requirements.

In open-field farming, tasks such as seeding, spraying, and harvesting require path planning that achieves complete coverage with minimal overlap and efficient headland turning. Navigation is dominated by coverage-route generation from field boundaries or pre-defined waylines, typically supported by RTK-GNSS guidance. Key requirements include coverage completeness, minimisation of overlap and skips, and efficient headland manoeuvres while avoiding obstacles such as trees, poles, and irrigation infrastructure.

In orchards and vineyards, navigation occurs in semi-structured environments characterised by narrow alleys, permanent tree rows, uneven terrain, and frequent occlusions. Typical tasks include precision spraying, mechanical pruning, mowing, and harvesting. Compared with open fields, the planning problem is more strongly constrained by collision risk, slope safety, and tree-row following under degraded GNSS reception. Accordingly, many studies adopt LiDAR- or vision-based row detection, trunk/canopy modelling, and multi-sensor fusion to maintain safe clearances to trees and terrain while satisfying implement trajectory requirements.

In protected cultivation (e.g., greenhouses, plant factories, or high-tunnel systems), the environment is spatially constrained yet geometrically regular, with fixed corridors, benches, or tracks. Navigation tasks include intra-row transport, crop inspection, targeted spraying, and harvesting. These tasks demand precise short-range manoeuvring, frequent docking or turning in tight spaces, and safe human–robot coexistence. Consequently, ultra-wideband (UWB), visual SLAM, LiDAR odometry, and quick response (QR) code or marker-based localisation are widely used to provide robust positioning and obstacle avoidance where GNSS is unavailable.

The mixed or "other" scenario category includes farm-road transport, field logistics, livestock feeding, and inspection operations that move between fields and facilities. Tasks range from global route planning on

road networks to local manoeuvring around animals or equipment. These settings often combine long-distance transport with docking or reversing in confined spaces and must handle dynamic obstacles such as vehicles, workers, and animals.

Across scenarios, navigation can be decomposed into three tightly coupled layers: (i) global task allocation and coverage planning at the farm/field scale; (ii) local path generation and re-planning that accounts for obstacles, curvature, and terrain; and (iii) path tracking and motion control at the vehicle level. The following subsections describe how platforms and information-acquisition modes are combined to address these layered requirements.

3.2. Agricultural ground machinery and typical platforms

The reviewed studies cover both retrofitted conventional machinery and dedicated UGVs. Despite diversity in size and actuation, these platforms share common requirements, including robust localisation, safe human–robot interaction, and reliable integration with task-specific implements.

Large wheeled tractors and sprayers dominate open-field operations and typically rely on RTK-GNSS and IMUs for precise guidance. Most use Ackermann steering with mounted or trailed implements and are retrofitted with RTK-GNSS receivers, IMUs, and, in some cases, cameras or LiDAR. Commercial auto-steering systems build on these configurations to deliver centimetre-level path tracking and automated headland turns for seeding, spraying, and harvesting. In research prototypes, additional sensors (e.g., row cameras or three-dimensional (3D) LiDAR) are often integrated to enable obstacle-aware coverage planning or terrain-adaptive control.

For small and medium-sized robots, tracked skid-steered UGVs and compact four-wheel-steered platforms are widely used in both arable and speciality crops. Tracked platforms provide strong traction and manoeuvrability on soft or uneven ground, supporting inter-row weeding, precision spraying, and in-row sensing. Four-wheel steering enables tight turning radii and lateral movements, which are advantageous in orchards and vineyards with complex headlands. These robots typically carry multi-sensor payloads combining GNSS, IMU, wheel odometry, stereo or red–green–blue–depth (RGB-D) cameras, and two-dimensional (2D)/3D LiDAR, supported by IPCs or embedded graphics processing unit (GPU) platforms for real-time processing.

In protected cultivation, lightweight wheeled or rail-guided robots are common. Compact size and limited payload capacity constrain sensor and computing choices, but also facilitate high-precision manoeuvres in narrow aisles. Typical configurations integrate UWB or visual markers for global localisation, RGB-D cameras or 2D LiDAR for obstacle detection, and on-board IPCs or Jetson modules for perception and control. Tasks such as spraying, picking, and transport often require repeated docking at workstations or benches, placing stringent demands on short-range positioning accuracy.

Specialised platforms have also been developed for specific tasks, including harvesting platforms, automated carriers for box transport, and multi-functional robots integrating spraying, weeding, and scouting modules. These systems emphasise tight coupling between vehicle motion and implement operation, together with safe human–machine interaction.

To compare systems systematically, the next subsection introduces a classification based on the environmental information used for navigation and the machinery operation mode.

3.3. Two-dimensional classification of information sources and operation modes

Building on the data extraction in Section 2.5 and the analytical framework in Section 2.6, each study was coded along two orthogonal dimensions (Fig. 2). The first dimension—information source for navigation—differentiates methods by the dominant information driving path generation and online updates. Static-information-based methods rely on a priori information such as field boundaries, headland paths, waypoints, digital elevation models (DEMs), or prescription/task maps. Routes are planned offline and executed online, typically using RTK-GNSS and an IMU for localisation, without local perception directly influencing path generation. Real-time-information-based methods rely on online perception for local planning and dynamic obstacle avoidance. Sensors such as cameras, LiDAR, radar, and ultrasonic devices detect crop rows, road edges, and obstacles to support on-the-fly path generation, re-planning, or target alignment. Hybrid-information methods integrate static maps with real-time sensing in hierarchical architectures. Common patterns include generating an offline global coverage plan and refining it online using crop-row or obstacle perception, or combining SLAM-based maps with explicit prior maps or guidance constraints (e.g., field boundaries and headland paths).

The second dimension—operation mode—distinguishes single-vehicle systems from multi-vehicle cooperative systems. In single-vehicle planning, autonomy is confined to an individual machine, and interactions with other machines are handled implicitly (e.g., via traffic rules, operator supervision, or fixed schedules). In multi-vehicle cooperative planning, explicit spatio-temporal coordination is required among fleets of machines or implements. Planning is therefore coupled with task allocation, conflict resolution, spacing or formation control, and may require communication and shared mapping.

Crossing these dimensions yields six conceptual categories: static-single, static-multi, real-time-single, real-time-multi, hybrid-single, and hybrid-multi. As detailed in Section 2.6, studies were assigned using explicit decision rules for dominant information source and for the presence of cooperative mechanisms.

Sections 4 and 5 use this framework to organise the technical synthesis of path planning methods under static, real-time, and hybrid information. Section 3.4 first provides a quantitative overview of how the 210 included studies distribute across years, scenarios, and the six categories.

3.4. Quantitative landscape of the literature

To complement the narrative synthesis, we analysed the coded dataset (Section 2.5) using descriptive statistics consistent with the framework in Section 2.6. In total, 210 studies were included. For the cross-tabulation in Table 1, we focused on 185 primary research articles that explicitly reported both a dominant information type and an operation mode, excluding review papers and simulation-only studies. These distributions reveal temporal trends and structural gaps that motivate the detailed discussion in Sections 4 and 5.

(i) Temporal trends and information sources. Fig. 5 shows that publications on autonomous navigation for agricultural ground machinery have increased markedly since 2019. Real-time-information-based methods account for the largest share in most years, reflecting widespread adoption of on-board sensing and local perception. Static-information-based approaches remain important, particularly for large-scale coverage planning in open fields. Hybrid-information methods appear later but increase steadily, indicating a shift towards architectures that combine global priors (e.g., field maps and task maps) with local sensing for refinement and obstacle avoidance. Overall, static maps remain a backbone for large-scale operations, whereas hybrid designs are increasingly adopted as a compromise between robustness and flexibility as sensing and computing capabilities improve.

(ii) Scenario-dependent operation modes. Fig. 6 summarises operation modes across the four scenarios in Section 3.1. The figure provides a scenario-level comparison of single- and multi-vehicle navigation, highlighting how deployment patterns differ among

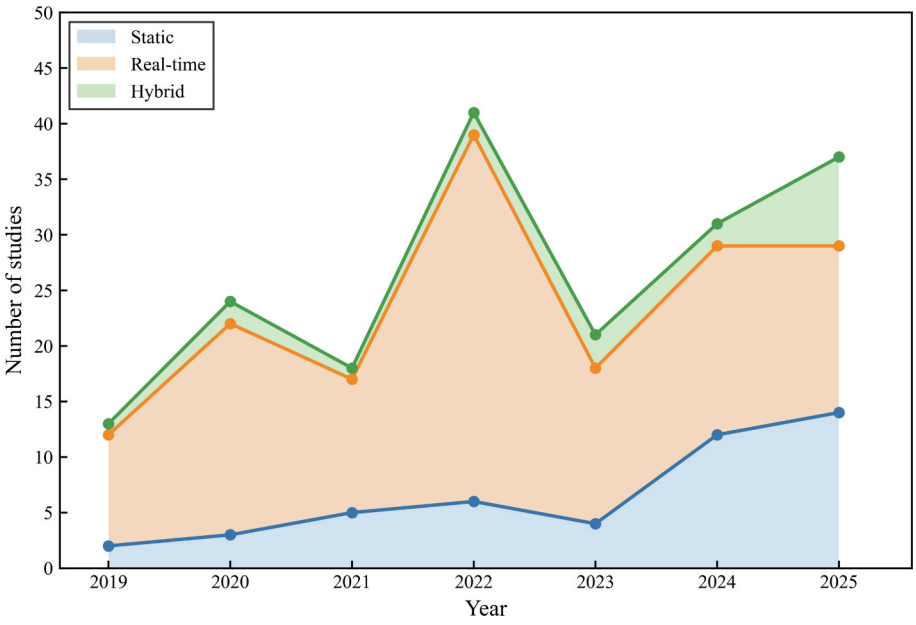


Fig. 5. Number of studies per year by information type (static, real-time, hybrid).

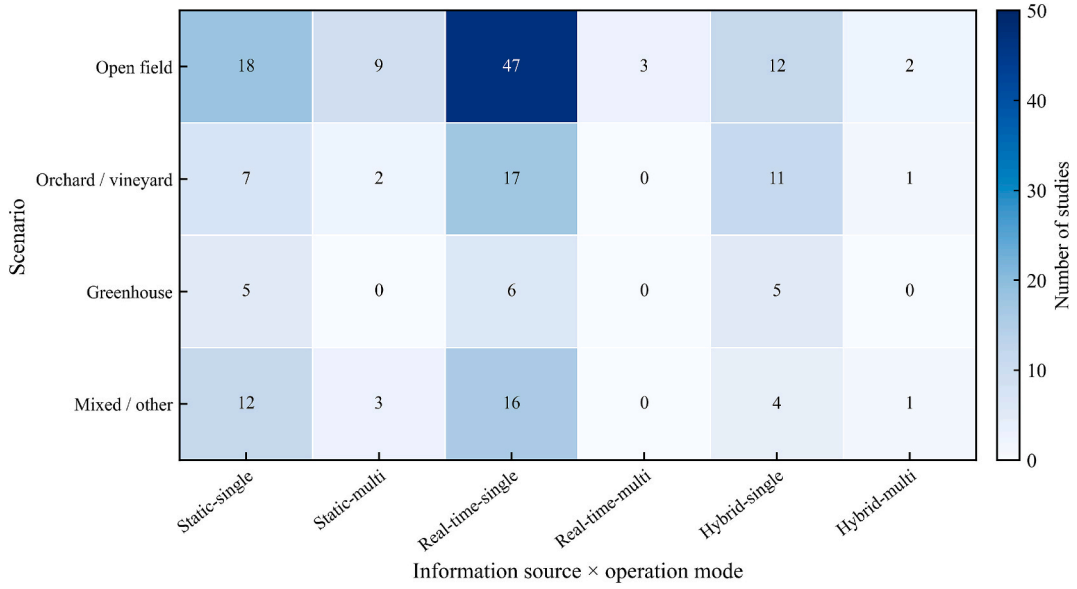


Fig. 6. Scenario-dependent distribution of operation modes in the reviewed studies.

open fields, orchards/vineyards, protected cultivation, and mixed farm environments. Single-vehicle navigation dominates in all scenarios, particularly in orchards and protected cultivation where tight spaces and safety constraints favour conservative deployment. In open-field operations, a non-negligible number of studies explore multi-vehicle cooperation, mainly in coverage and logistics tasks where collision risk and communication constraints are more manageable. The mixed-scenario group is again dominated by single-vehicle deployments, although some studies propose fleet management or convoy strategies where cooperative navigation could further improve efficiency.

(iii) Information type vs operation mode. While Fig. 6 focuses on scenario-dependent operation modes, Table 1 provides an aggregated numerical cross-tabulation of information types and operation modes. This table directly supports the six-category framework defined in Section 2.6 by reporting the exact study counts for static-single, static-multi, real-time-single, real-time-multi, hybrid-single, and hybrid-multi systems. Static-single and real-time-single together account for most of the 185 primary studies, indicating that autonomy remains predominantly implemented at the single-vehicle level. Multi-vehicle studies are fewer and are skewed towards static-multi and hybrid-multi configurations, where global task maps or pre-computed routes are combined with limited local perception and relatively simple coordination rules. Only a small number of studies fall into the real-time-multi category, reflecting the difficulty of tightly coupling multi-vehicle coordination with perception-driven local planning under realistic communication constraints. Hybrid-single often serves as a transitional architecture towards hybrid-multi deployments.

Taken together, Fig. 6 and Table 1 provide complementary evidence:

Fig. 6 highlights scenario-dependent deployment patterns, whereas Table 1 quantifies the overall structural imbalance among the six information-source–operation-mode categories. To avoid overloading the main text with listing-style tables, the detailed study-level records were reorganised as Appendix Tables A1-A9. In the main text, these records are synthesised through representative KPI extraction and graphical summaries: Table 2 and Fig. 7 summarise reported performance indicators, Fig. 8 shows the scenario-dependent distribution of algorithm categories, and Fig. 9 summarises sensor and information-source configurations across scenarios.

(iv) Representative KPI synthesis and performance visualisation. To complement the categorical statistics and to provide a more direct comparison of reported navigation performance, representative KPIs were further extracted from typical studies across open-field, orchard/vineyard, greenhouse, and mixed agricultural scenarios (Table 2). The extracted indicators include lateral deviation, localisation or positioning error, heading or angular error, processing latency or frame rate, vehicle speed or operating condition, sensor configuration, and representative algorithm or method.

Localisation, path extraction, local planning, and path tracking jointly determine the deployable performance of agricultural ground machinery under heterogeneous field conditions, such as canopy occlusion, GNSS degradation, narrow greenhouse aisles, and dynamic obstacles. To provide a structured performance overview, Table 2 summarises representative algorithms, scenarios, sensor configurations, error ranges, angular errors, processing latency, and frame rate. Representative KPI records from Table 2 were used to construct Fig. 7. Directly comparable accuracy-related values, including lateral deviation, localisation error, and positioning error, were converted to centimetres for Fig. 7(a), whereas processing latency and FPS-derived latency were converted to ms frame $^{-1}$ for Fig. 7(b). Studies reporting only angular error, percentage error, detection accuracy, or task-level indicators were not plotted in Fig. 7(a). Appendix Table A10 separately summarises reported performance improvements relative to baseline methods, including reductions in lateral deviation, heading deviation, path cost, operation time, and coverage-related indicators.

To make these heterogeneous KPI records more visually interpretable, the most directly comparable accuracy-related and real-time indicators from Table 2 were further plotted in Fig. 7.

The extracted KPIs show that centimetre-level tracking or localisation accuracy is frequently reported in structured open-field, orchard, and protected-cultivation settings, particularly when RTK-GNSS, UWB, LiDAR, vision, or multi-sensor fusion is used. However, wider error ranges are observed in more complex orchard, vineyard, paddy-field, and mixed agricultural scenarios because of canopy occlusion, uneven terrain, irregular crop-row geometry, GNSS degradation, and dynamic disturbances. Because platform scale, vehicle speed, crop type, terrain condition, sensing configuration, and validation protocol vary substantially among studies, the values in Table 2 and Fig. 7 should be interpreted as representative KPI ranges rather than as a controlled benchmark or *meta*-analysis.

Beyond the KPI-level comparison in Fig. 7, Fig. 8 summarises how representative algorithm categories vary across agricultural scenarios. Open-field studies involve a broad mixture of coverage/decomposition planning, graph search, tracking control, and metaheuristic optimisation, reflecting the importance of complete coverage, headland manoeuvres, and large-scale route efficiency. Orchard and vineyard studies show stronger use of vision/image-processing, deep learning, SLAM/localisation, and tracking-control methods, consistent with canopy occlusion, degraded GNSS reception, and row-structure constraints. Because one study may include multiple algorithmic modules, the values in Fig. 8 represent category occurrences rather than mutually exclusive study counts.

Fig. 9 summarises how sensor and information-source configurations differ across scenarios. Open-field studies are dominated by GNSS/RTK, IMU/INS, RGB-camera-based perception, UAV imagery, and prior-map information. In orchards and vineyards, RGB vision, LiDAR, GNSS/RTK, and IMU/INS appear more frequently in combination, indicating the need for localisation redundancy under canopy occlusion and intermittent satellite positioning. Greenhouse and protected-cultivation studies show a smaller but more diverse sensor profile, including UWB, RGB-D/stereo vision, LiDAR, and prior-map information. Because one study may use multiple sensors or information sources, the values in Fig. 9 represent sensor-category occurrences rather than mutually exclusive study counts.

3.5. *Applications and commercialisation of navigation systems*

Commercial navigation solutions provide a reference for assessing the maturity and deployability of research prototypes. Fig. 10 illustrates representative examples, including (a) John Deere AutoTrac®, (b) Case IH AFS™, and (c) DENSO & Certhon Artemy®.

Commercial systems such as John Deere AutoTrac® and Case IH AFS™ exemplify mature static-single deployments, relying on RTK-GNSS, IMU, and wheel-encoder fusion for centimetre-level path tracking in open fields with an emphasis on coverage efficiency and task logging. Although local obstacle avoidance is typically limited, their long-term commercial success demonstrates that static, map-based navigation can deliver substantial gains in operational efficiency, fuel economy, and operator comfort in structured environments.

In contrast, greenhouse robots such as DENSO & Certhon's Artemy® reflect real-time-single or hybrid-single deployments. Operating in GNSS-denied and cluttered environments, these platforms integrate UWB, visual SLAM, or LiDAR-based localisation with dense perception for crop and obstacle detection. Their navigation focuses on precise docking, short-range obstacle avoidance, and safe human–robot collaboration, consistent with protected-cultivation requirements (Section 3.1).

Across these commercial examples, three common trajectories emerge. First, multi-sensor fusion (GNSS, IMU, LiDAR, vision, and UWB) has become standard practice to stabilise localisation under variable conditions. Second, path planning is tightly coupled to machinery kinematics and farm-management workflows rather than implemented as a stand-alone generic module. Third, most commercial systems remain single-vehicle at the navigation level, even when deployed in fleets; cooperative planning is typically handled by external management systems or human operators rather than by explicit multi-vehicle navigation algorithms.

By mapping commercial systems into the same two-dimensional framework as the research literature, this section links navigation problems, scenarios, classification, and deployment considerations. Sections 4 and 5 build on this foundation to compare technical routes for static- and real-time-information-based planning and to identify bottlenecks and future directions for agricultural navigation systems.

4. Single-vehicle navigation: Environmental perception and path planning

Single-vehicle navigation remains the most mature and extensively studied paradigm for autonomous agricultural ground machinery, accounting for over 70% of the studies included in this review (Section 3.4). As shown in Fig. 6, most single-vehicle systems target open-field and orchard operations and rely predominantly on static or real-time information, whereas hybrid-information settings remain less represented and multi-vehicle cooperation is comparatively scarce. This section therefore examines how environmental information is perceived and exploited to support path planning and navigation for an individual machine. The discussion has been organised into four layers of increasing system complexity: static-information-based global planning (Section 4.1), real-time information sources and perception (Section

Table 2
Representative KPIs reported in typical agricultural ground-machinery navigation and path-planning studies.

Algorithm / architecture	Application scenario and platform	Core sensor configuration	Lateral / positioning error	Heading / angular error	Processing latency / FPS	Vehicle speed / operating condition	Reference
NPENet single-stage path-line extraction network	Orchard mobile robot	Machine vision	NR	NR	10.1 ms/frame on Jetson Xavier	NR	Liu et al. (2025b)
RTK-BDS/IMU integrated navigation with Pure Pursuit	Orchard spraying robot	RTK-BDS + IMU	Maximum offset error < 10.77 cm; average offset ≤ 3.55 cm; lateral RMSE = 1.19 cm	Position error $<$ 2.49 cm and heading error $< 2^\circ$ during straight movement; static heading error $< 0.3^\circ$	NR	0.4 m/s	Zhang et al. (2022b)
Midpoint Hough transform	Greenhouse tomato–cucumber spraying robot	RGB vision system	NR; path-fitting angular error was reported instead of lateral deviation	Maximum path-fitting error = 0.5°; average accuracy improvement over least-squares fitting = 5.1°	7.13 ms/frame	Greenhouse row-following test	Chen et al. (2020a)
Improved edge detection + Otsu thresholding	Cotton-field tractor / crop-row navigation	Machine vision	SD of lateral deviation = 2.0 cm	SD of heading deviation = 0.57°	6.63 ms/frame for cotton; 6.97 ms/frame for maize; 6.72 ms/frame for soybean	Seedling-stage crop-row detection	Liang et al. (2022)
Patch-based convolutional neural network (CNN)	Citrus-orchard autonomous rover / path detection	IP camera	NR	Average angular error = 7.8°	111 ms/frame	Patch size 96×96 gave the best performance	Kim et al. (2020)
Improved DeepLabv3 + with DR-ASPP	Field headland turning agricultural vehicle	RGB-D multimodal vision	Maximum row-change deviation = 0.09 *m*	NR	23.1 FPS	Headland turning / crop-bank detection	Wu et al. (2025)
ESANet RGB-D semantic segmentation	Irregular small-field single-side navigation	RGB-D depth camera	Mean deviation $<$ 0.188 *m*	NR	23.80 FPS	Irregular field / cape navigation	Li et al. (2024a)
CGNet with self-attention	Farmland road-intersection UGV	Machine vision	Mean lateral deviation $<$ 4% of road width / centreline reference	NR	104 FPS	Mountainous farmland road intersections	Li and Wu (2025)
Faster U-Net	Greenhouse multi-crop ridge extraction	Machine vision	NR; ridge-line angular fitting errors were reported	Average ridge-line angular errors: maize 0.624°, tomato 0.556°, cucumber 0.526°, wheat 0.999°	22.32 FPS on a single-core CPU	Multi-crop ridge extraction	Li et al., (2022c)
Improved SIFT-based feature matching and depth recovery	Unstructured agricultural road visual navigation	Binocular stereo vision	Not a lateral-deviation metric; depth-recovery relative error ranged from -7.17% to 2.97%; matching accuracy = 96.48%; matching accuracy $>$ 88% across 60 images	NR	5.6 s for the improved SIFT method; average 5.0–6.5 s under different weather and road conditions	Offline / near-real-time visual depth recovery	Yao et al. (2019)
Particle filter (PF) with laser-beam model	Apple-orchard four-wheel robot	2D LiDAR + IMU + wheel encoders + RTK-GNSS	Lateral RMSE = 25.0 mm at 2.0 m/s; 50.0 mm at approximately 0.87 m/s	Angular deviations were in the same range as KF	NR	2.0 m/s and approximately 0.87 m/s	Blok et al. (2019)
PSO-enhanced fuzzy Stanley controller	Field combine harvester / full-field path tracking	GNSS positioning system	Maximum lateral error reduced from 0.32 m to 0.03 m in the reported comparison; preliminary field maximum tracking error = 0.63 *m*	NR	NR	Adaptive vehicle speed / actuator saturation conditions	Sun et al. (2022b)
Dynamic look-ahead Pure Pursuit	Cattle-barn feed-pushing robot	UWB positioning	RMSE = 4.22 cm	NR	NR	Low speed: 6–10 m/min	Chen et al. (2024)
Improved A* search + fuzzy Dynamic Window Approach (DWA)	Greenhouse–orchard mobile robot	LiDAR + vision	Maximum lateral deviation = 11.20 cm	Maximum heading deviation = 13°	NR	NR	Guo et al. (2024a)
Improved A* search + SVR motion model	Orchard tracked platform	3D LiDAR point cloud	Mean maximum lateral and longitudinal deviations = 6.90 cm	NR	NR	Multiple track-speed combinations	Liu et al. (2025c)

(*continued on next page*)

Table 2 (*continued*)

Algorithm / architecture	Application scenario and platform	Core sensor configuration	Lateral / positioning error	Heading / angular error	Processing latency / FPS	Vehicle speed / operating condition	Reference
			and 9.88 cm, respectively				
Normal Distributions Transform (NDT) + Iterative Closest Point (ICP) point-cloud registration SLAM	Orchard spraying robot	3D LiDAR + IMU	Maximum positioning error = 7.04 cm	Mean heading deviation < 8°	NR	1.2 m/s	Jiang et al. (2024)
Dynamic GNSS/INS/LiDAR switching	GNSS-occluded orchard autonomous equipment	RTK-GNSS + IMU + LiDAR	Mean lateral error = 0.10 *m*	NR	NR	Positioning error reduced by 1.6 m compared with the degraded baseline	Li et al. (2024d)
YOLOv5 + LQR control	Fruit-tree recognition and navigation	Machine vision	Steady-state lateral error = 0.0102 *m*	Steady-state heading error = 0.0253 rad (approximately 1.45°)	NR	Initial speed = 0.5 m/s	Xie et al. (2024)
Vision-based crop-row detection with GNSS and compass integration	Rice-field weeding robot	GNSS + compass + machine vision	Mean deviation from the ideal path = 45.9 mm	Heading compensation accuracy < 2.5° under low weed density	NR	Rice-field operation under variable weed density	Kanagasingham et al. (2020)
Medial-axis-based machine-vision navigation	Orchard tracked robot	RGB camera / machine vision	Maximum tracking error = 14.6 mm; SD = 6.8 mm; lateral RMSE = 45.3 mm	Heading and offset used as fuzzy-controller inputs; angular error NR	NR	Orchard field experiment	Opiyo et al. (2021)
Extended Kalman filter (EKF)-based navigation using LiDAR and gyroscope	Rubber-tree tapping robot / forest row navigation	Low-cost 2D LiDAR + gyroscope	RMS lateral error < 10.32 cm; average stopping error = 12.08 cm	Heading error analysed but exact value NR in the extracted summary	NR	Initial speed = 0.3 m/s	Zhang et al. (2019)
Improved random sample consensus (RANSAC) navigation-line detection	Rice/wheat intelligent harvester	Vision system; Jetson Xavier	NR; navigation-line detection success rate = 94.6%	NR	57.1 ms/frame	Harvest boundary detection	Li et al. (2020)
Stereo-vision crop-edge detection with dynamic ROI	Multi-crop combine harvester	Binocular/ stereo camera	NR; crop-edge detection accuracy > 98% for rice and rape, >94% for maize	NR	49 ms/frame	Rice, rape, and maize harvesting edge detection	Luo et al. (2022)

Note: NR indicates that the corresponding metric was not reported in the original study. Several studies reported perception- or path-extraction metrics rather than direct vehicle-tracking errors; these entries were retained because real-time path extraction is an upstream bottleneck for agricultural navigation. Values were extracted as reported by the original authors and converted only when unit conversion was straightforward, such as m to cm or FPS to ms frame -1. The accuracy-related indicators in Table 2 support Fig. 7(a), whereas latency and frame-rate indicators in Table 2 support Fig. 7(b). Appendix Table A10 separately summarises reported performance improvements relative to baseline methods. Because platforms, crops, speeds, sensing configurations, and validation protocols differ substantially across studies, the values should be interpreted as representative KPI ranges rather than as a controlled benchmark or *meta*-analysis.

4.2), real-time local path planning (Section 4.3), and hybrid-information-based planning and control that integrate prior maps with online perception and SLAM (Section 4.4). For each layer, we highlight dominant modelling strategies, algorithmic trends, validation practices, and key technical bottlenecks, thereby linking the scenario-aware framework in Section 2.6 to the quantitative landscape summarised in Section 3.

4.1. Static-information-based global path planning for single-vehicle systems

Within the two-dimensional framework in Section 2.6, static-information-based single-vehicle navigation represents both the earliest and still a substantial class of agricultural navigation systems. These approaches assume that the operational geometry—field boundaries, obstacles, access routes, and elevation—is available a priori and remains unchanged during execution. Consistent with Section 3.4, static-single systems are most prevalent in studies of complete coverage in open fields and structured orchards, particularly for sowing, spraying, mowing, and harvesting.

Static-information-based planners typically follow a three-stage pipeline: (i) environment modelling using satellite imagery, UAV orthomosaics, or pre-surveyed LiDAR/GNSS data; (ii) formulation of optimisation criteria, commonly including coverage efficiency, travel distance, turning effort, smoothness, or energy consumption; and (iii) application of global search or optimisation algorithms to generate executable trajectories. Here, we distinguish complete coverage path planning (CCPP), which enforces full traversal of the target area (Section 4.1.1), from multi-objective complete coverage path planning (MO-CCPP), which seeks a compromise among multiple criteria and may relax strict coverage constraints (Section 4.1.2).

4.1.1. Global complete coverage path planning

In CCPP, the planner generates trajectories that traverse all reachable areas with minimal overlap and omission. Early studies were largely geometry-driven. Lu et al. (2020) and Martelloni et al. (2019)

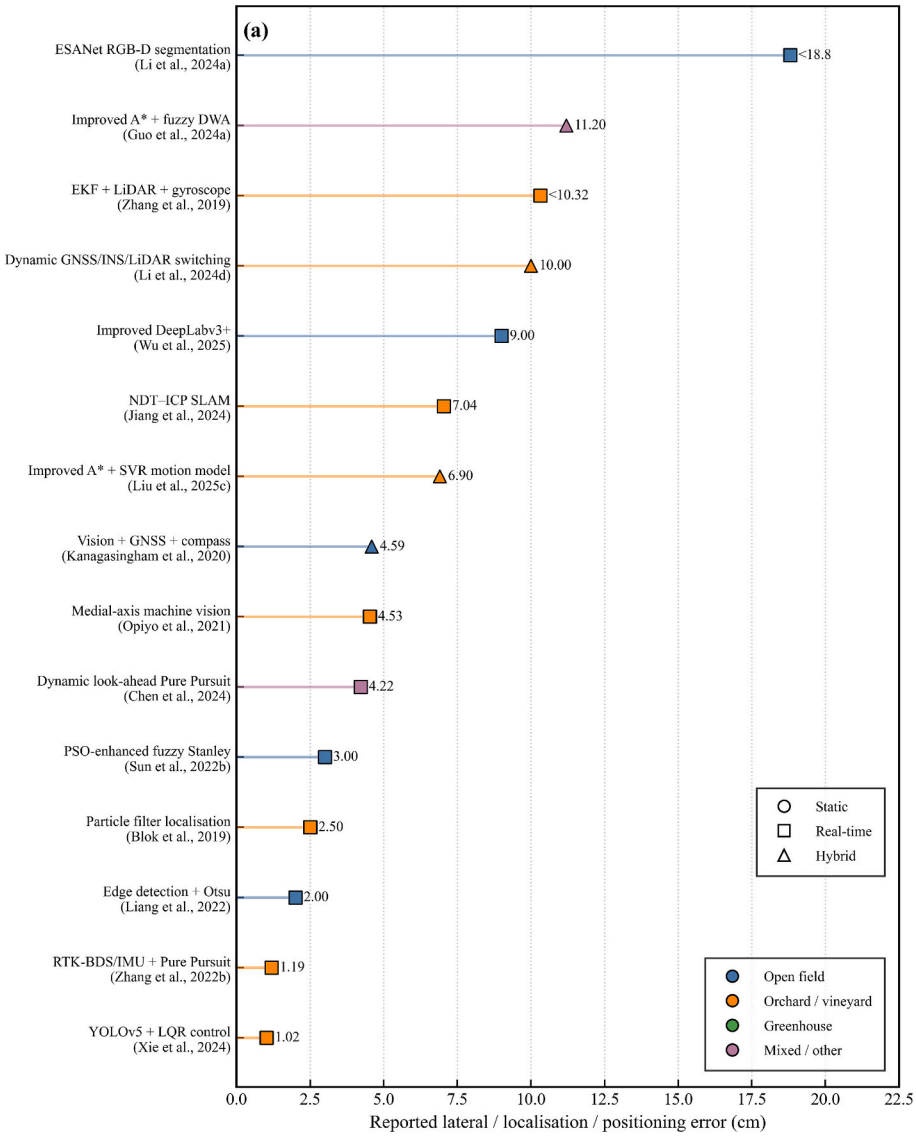


Fig. 7. Representative KPI comparison of agricultural navigation methods: (a) accuracy-related indicators; (b) real-time processing indicators.

derived polygonal field representations from satellite imagery and RTK-GNSS trajectories, respectively, and showed that plot geometry and boundary regularity directly influence coverage efficiency. Mazzia et al. (2021) introduced deep-learning-based waypoint extraction from aerial imagery, bridging the gap between raw imagery and executable global routes.

More recent work has moved towards composite static priors that combine geometry with semantics and operational constraints. The detailed CCPP records in Appendix Table A1 support this transition from geometry-driven coverage planning towards semantics-aware global planning. For example, Lin et al. (2025) fused UAV imagery with YOLO-based segmentation to identify traversable regions in pineapple fields, whereas Wang et al. (2024b) embedded operation semantics, such as blade-lifting points, into cotton-field coverage routes and reported coverage rates above 98%. These studies correspond to the coverage/decomposition, graph-search, metaheuristic-optimisation, and deep-learning categories summarised in Fig. 8.

Across the CCPP literature, five recurring algorithmic trends can be identified:

(1) Refinement of graph-search heuristics. Weighted and bidirectional A* variants combined with curve smoothing are increasingly used to improve computational efficiency and path smoothness (Li et al., 2022a; Guo et al., 2024a; Chatzisavvas et al., 2024). Field experiments in greenhouses and orchards commonly report shorter planning times and reduced lateral tracking errors (often < 0.12 m) relative to baseline A*.

(2) Hierarchical global–local coordination. Improved A* or D* Lite planners are often coupled with local planners/controllers (e.g., fuzzy-DWA and Hybrid A*) to handle narrow orchard aisles and irregular headlands (Wang et al., 2024d; Jiang et al., 2025b), with reported reductions in turning angles and pose deviations exceeding 30% in representative cases.

(3) Coverage as combinatorial optimisation. By formulating coverage planning as travelling-salesperson or region-sequencing problems, studies such as Ait et al. (2023) and Wang et al. (2025) report 11–20% time savings while maintaining coverage rates above 97%.

(4) Acceleration for large-scale environments. Meta-heuristics and jump-point-search-based strategies (Zhang et al., 2025c; Guo

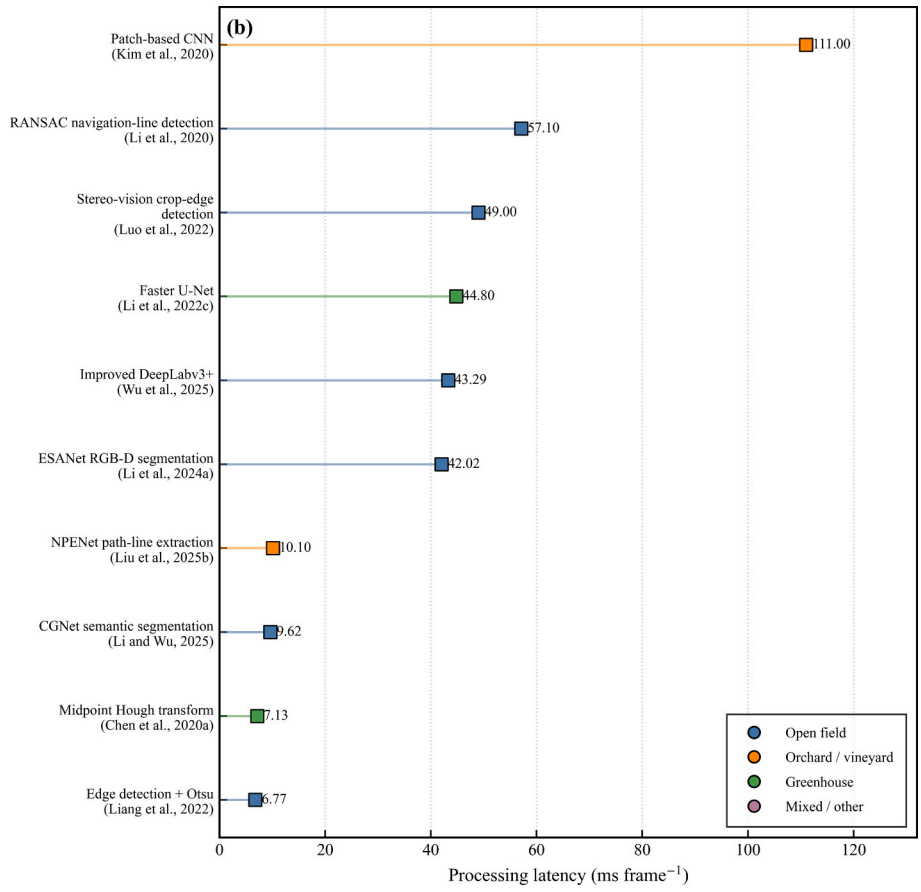


Fig. 7. (*continued*).

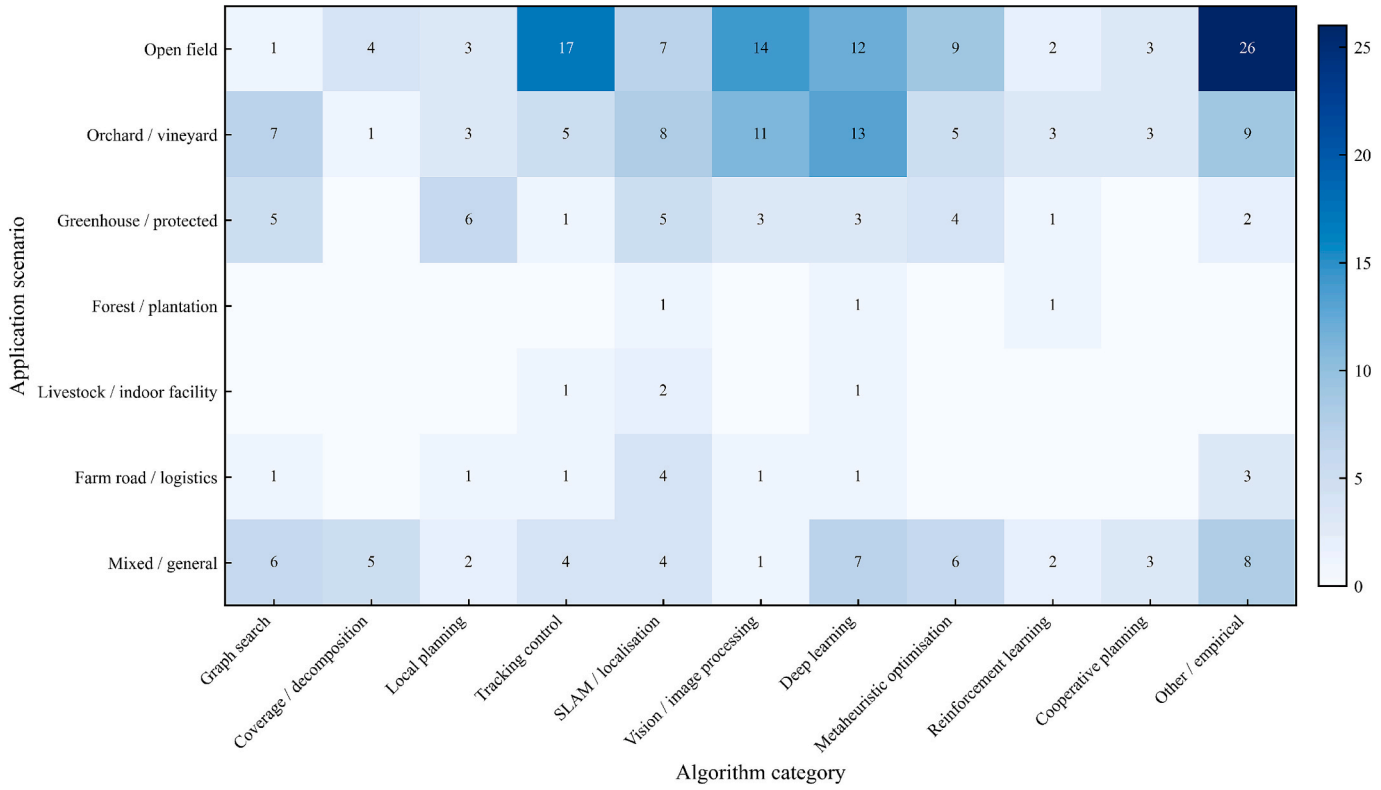


Fig. 8. Scenario-dependent distribution of representative navigation and path-planning algorithm categories.

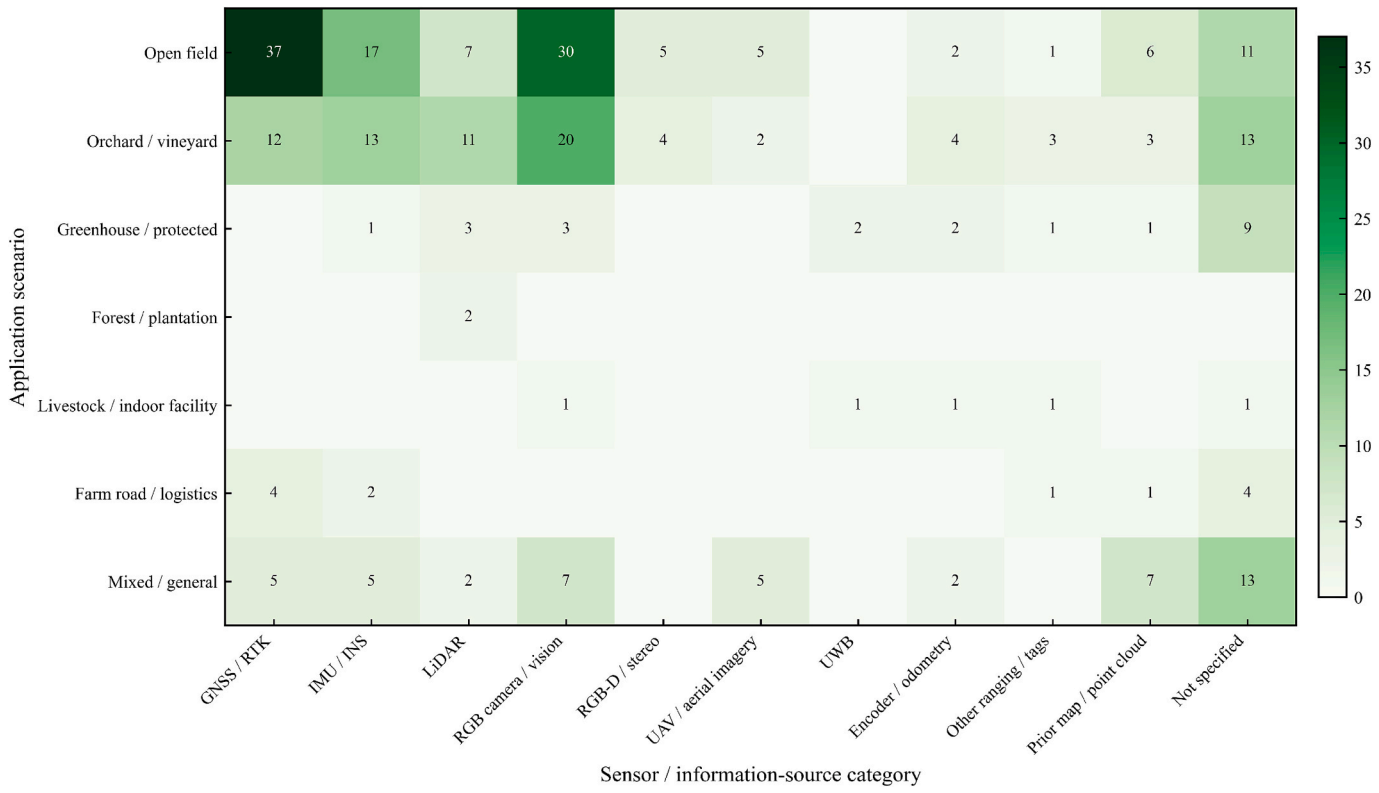


Fig. 9. Sensor and information-source configurations across agricultural navigation scenarios.



(a) (b) (c)

Fig. 10. Representative commercial agricultural navigation systems: (a) John Deere AutoTrac® system; (b) Case IH AFS™ navigation system; (c) DENSO & Certhon Artemy® greenhouse harvesting robot.

et al., 2025) explicitly target computational scalability, which is critical for large static grids and facility-like layouts.

(5) Agriculture-aware cost modelling. Time-dependent spatial planning and energy-aware heuristics (Zhang et al., 2024c; Chen et al., 2025a) incorporate turning energy, implement dynamics, and terrain constraints to improve feasibility under realistic operating conditions.

Despite these advances, most CCPP studies still rely on idealised static maps and short-term experiments on single plots. As indicated by the validation statistics in Section 3.4, multi-season or cross-farm evaluations remain rare, and agronomic side effects (e.g., soil compaction and long-term traffic patterns) are seldom assessed.

4.1.2. *Global multi-objective overlay path planning*

The detailed MO-CCPP records in Appendix Table A2 show that hierarchical decomposition and evolutionary or swarm-based optimisation remain dominant for multi-objective coverage planning. Yang et al. (2024) combined A*-based region traversal with genetic-algorithm sequencing, whereas Kong et al. (2024) and Xie et al. (2024) integrated Bézier-curve smoothing with hybrid ant-colony strategies to improve kinematic feasibility. These studies are reflected in Fig. 8 through the graph-search, coverage/decomposition, and metaheuristic-optimisation categories.

Energy-aware planning has emerged as a distinctive line of work. Wei (2021) trained deep neural networks to estimate energy-cost maps, enabling coverage routes that respect battery constraints. Xie et al.

(2025) found that integrating slope information into multi-robot task allocation reduced energy consumption by 4.5–10.3% for electric weeding machines operating in hilly orchards. In contrast to CCPP, MO-CPP studies exhibit a wider range of objectives and algorithm families, yet their evaluation practices tend to be more fragmented. Many experiments remain simulation-based or confined to specific crops and terrains, limiting generalisability. These observations motivate the need for benchmark scenarios and shared datasets that explicitly link coverage quality, energy efficiency, and agronomic outcomes, as discussed further in Section 6.

4.2. Real-time information sources and perception methods for single-vehicle navigation

While static-information-based planning is prevalent in early-stage system design, the quantitative analysis in Section 3.4 indicates that a growing share of studies published after 2021 rely primarily on real-time information for single-vehicle navigation, particularly in orchards and protected cultivation. As shown in Fig. 6 and Table 1, real-time-single systems form the largest category in these semi-structured environments, reflecting the need to cope with GNSS degradation, vegetation occlusion, variable illumination, and dynamic obstacles.

In real-time navigation, path planning is not governed solely by a fixed map; instead, the navigation stack must continuously maintain and update a local representation of the environment from on-board sensing. Consequently, two closely interdependent functions emerge as critical: (i) real-time localisation, which estimates vehicle pose under degraded or intermittent positioning conditions, and (ii) environmental perception, which extracts navigable structures and obstacles to support local path generation and re-planning.

4.2.1. Real-time localisation

RTK-GNSS remains the primary source of absolute positioning in open fields and relatively unobstructed arable land. The detailed real-time information-acquisition records in Appendix Table A3 show that, under favourable open-field conditions, centimetre-level accuracy can be obtained using loosely or tightly coupled GNSS–IMU fusion. This pattern is also visible in Fig. 9, where GNSS/RTK and IMU/INS form major sensor categories in open-field navigation studies. (Si et al., 2019; Fue et al., 2020; Jing et al., 2023).

To bridge GNSS outages and maintain pose continuity, most systems adopt multi-sensor fusion. Common configurations integrate GNSS with IMUs and wheel encoders, and may additionally incorporate crop-row detectors, using extended, unscented, or cubature KF and related Bayesian estimators (Blok et al., 2019; Li et al., 2021b; Jing et al., 2023). In orchard and greenhouse settings, 2D/3D LiDAR and monocular or RGB-D cameras are increasingly used for scan matching, visual odometry, or Monte Carlo localisation, enabling relative localisation in GNSS-denied environments (Tinchev et al., 2019; Guevara et al., 2021a; de Lima et al., 2023). Recent work further explores adaptive switching between GNSS/ inertial navigation system (INS) and LiDAR-based localisation to improve robustness under dense canopy conditions; reported lateral errors are on the order of 0.10 m in representative experiments (Li et al., 2024d).

Overall, Appendix Tables A3 and Fig. 9 indicate a transition from GNSS-centric pipelines towards heterogeneous sensor fusion. Filter-based approaches remain dominant because of their interpretability and moderate computational requirements, while LiDAR-, camera-, UWB-, and odometry-based localisation are increasingly used where GNSS becomes unreliable.

4.2.2. Environmental perception

Environmental perception provides the planner with structural and semantic information about the workspace, including crop rows, road boundaries, static obstacles, and dynamic agents (e.g., workers and livestock). Camera-based perception is widely adopted for row and road extraction owing to low cost and high information density. Classical pipelines based on colour indices, edge detection, and Hough transforms remain used in controlled conditions; however, deep-learning-based detection and segmentation have become increasingly prevalent since 2020, consistent with the growth of real-time-single studies in Table 1.

LiDAR is primarily employed for obstacle detection, clearance estimation, and terrain assessment. Point clouds are typically converted into local occupancy grids, elevation maps, or slope maps, which are then registered to GNSS or SLAM frames to support local planning and collision avoidance. In livestock facilities and protected cultivation, where illumination and occlusion are particularly challenging, infrared cameras, UWB anchors, and fiducial markers (e.g., DRI tags and April-Tags) are used to stabilise perception and localisation (Zhang et al., 2024a; Chen et al., 2024).

A recurring trend indicated by Appendix Table A3 and visually reinforced by Fig. 9 is the move towards multi-modal perception and localisation. Systems combining cameras, LiDAR, inertial sensing, and, in some cases, UWB or marker-based localisation report improved robustness to occlusion, illumination changes (Yao et al., 2024; Fujinaga, 2025), and GNSS degradation. This supports the broader shift from single-sensor navigation towards hybrid information architectures in complex agricultural environments.

4.2.3. Real-time processing and system constraints

Real-time perception and localisation impose stringent constraints on computing platforms and system latency. Most reported systems run on embedded controllers or IPC-class computers with update cycles on the order of 10–100 ms, which is often sufficient for low-speed field robots but can become a bottleneck for higher-speed operations including road transport or wide-boom spraying. Addressing these constraints, recent studies favour lightweight neural networks, efficient point-cloud processing, and modular software architectures, commonly built on Robot Operating System (ROS) or comparable middleware.

Despite these advances, Fig. 6 and Table 1 reveal an imbalance: real-time perception and localisation are predominantly exploited in single-vehicle deployments, whereas real-time information sharing and coordination across multiple vehicles is rarely implemented in comparable depth. The disparity between the real-time single-agent and real-time multi-agent categories necessitates the subsequent analysis of local planning and hybrid architectures (Sections 4.3–4.4), as well as the discussion of cooperative systems and evaluation requirements in Section 6.

4.3. Real-time-information-based local path planning for single-vehicle systems

Real-time information is central to the robustness of autonomous navigation for agricultural ground machinery because operational conditions can deviate rapidly from any pre-defined plan. Whereas global planning provides field-scale routes and task-level ordering, local path planning governs the short-horizon trajectory that the vehicle executes in the presence of local obstacles, row-structure variability, terrain irregularities, and actuator/kinematic constraints. In agricultural settings, this local layer is particularly consequential in semi-structured environments—such as orchards, vineyards, protected cultivation, and mixed farm traffic—where GNSS may be degraded and the navigable corridor is conditioned by vegetation, implements, and human activity.

This section synthesises recent advances in real-time-single local planning with a scenario-oriented view, focusing on (i) farm roads, (ii) crop rows, and (iii) row changes and headland manoeuvres. Across these scenarios, a common engineering pattern emerges: perception does not directly yield a trajectory; instead, it first produces an intermediate representation (e.g., centreline, free-space boundary, or occupancy/elevation grid), which is then consumed by a local planner and stabilised by a tracking controller. We hence emphasise how perception, representation, and planning are coupled, what design trade-offs are

reported, and which gaps remain for deployable systems.

4.3.1. Local path planning for farm roads

Farm roads connect fields, storage areas, and facilities, and thus serve as a transition environment between field operations and mixed-traffic transport. Although farm roads are typically wider than crop rows, they are less standardised than public roads: surfaces are uneven, boundaries can be weakly defined, and dynamic interactions with other vehicles, workers, and animals are frequent. Effective navigation accordingly requires real-time detection of the drivable corridor and local planning that adapts to terrain disturbances and transient obstacles.

Earlier approaches often relied on hand-crafted image processing and simple sensor fusion, for example extracting road edges from RGB imagery or range data and generating a centreline-following trajectory (Tian et al., 2022; Yao et al., 2019). These pipelines can be effective in constrained conditions but degrade under changing illumination, seasonal vegetation encroachment, dust, and partial occlusions—factors that are common on farms (Yang et al., 2022b). Recent work has increasingly adopted learning-based perception (e.g., semantic segmentation) combined with geometry-aware representations. In this design, deep networks provide dense traversability cues, while LiDAR or depth sensing supports obstacle detection and free-space estimation via occupancy grids. Local planners then optimise the trajectory for collision avoidance, curvature continuity, and tracking feasibility, rather than merely following a detected centreline.

A notable direction is the integration of prior information (e.g., GNSS-referenced farm maps or previously surveyed corridors) with online sensing to stabilise localisation and reduce perception burden. Representative studies (e.g., Li et al., 2024b; Wang et al., 2024c) demonstrate that combining map priors with real-time sensing can improve path smoothness and localisation continuity in dynamic farm-road environments. Methodologically, this trend can be interpreted as a transition from pure real-time-single to hybrid-single designs at the local-planning layer, even when the global route remains map-driven.

Looking forward, farm-road planning faces two recurring challenges: (i) robust free-space perception under adverse illumination and weather, and (ii) consistent risk-aware behaviour when encountering moving agents. Reinforcement learning (RL) and other learning-based decision policies are increasingly explored to handle rare events and uncertainty, but their deployment requires careful safety constraints and evaluation protocols. Furthermore, although this section focuses on single-vehicle systems, mixed-traffic farm roads naturally motivate future extensions towards cooperative interaction rules and fleet-aware scheduling, which remain limited in the current literature.

4.3.2. Local path planning for crop rows

Crop rows constitute the dominant navigational structure in arable fields, orchards, and vineyards, where machinery must operate in narrow corridors with strict crop-clearance constraints. Local planning in rows differs fundamentally from farm-road navigation: the feasible space is bounded by vegetation, the tolerance for lateral deviation is small, and occlusion and self-similarity are recurring. Consequently, most row-navigation pipelines adopt a “row-structure-first” representation, such as a row centreline or paired boundary lines, which then drives trajectory generation and tracking control.

Early systems often used GNSS-referenced row maps or predefined waylines, which are effective when row geometry is regular and obstacles are sparse.

Online row perception is increasingly required in practical deployments, as row continuity can be compromised by factors such as missing plants, lodged crops, variable canopy closure, and shadow patterns. The detailed crop-row extraction and navigation-line tracking records in Appendix Tables A4 and A5 show a progression from geometry-based extraction, such as least-squares fitting, Hough/RANSAC, and clustering, to learning-based segmentation and detection. This progression is reflected in Fig. 8, where vision/image-processing and deep-learning categories are widely represented in crop-row and orchard/vineyard navigation scenarios. It is also consistent with Fig. 9, which shows that RGB cameras, RGB-D/stereo vision, LiDAR, and GNSS/IMU combinations are frequently employed in row-guided navigation.

From an engineering perspective, two technical bottlenecks dominate row-based local planning. First, perception robustness remains sensitive to domain shifts (crop species, growth stage, canopy density, and lighting), which can lead to unstable centreline estimates and oscillatory control. Multi-sensor fusion—combining vision with LiDAR/depth, inertial sensing, and, when available, GNSS priors—has therefore become a common strategy to stabilise the row representation before planning. Second, planning and control are often evaluated separately, despite strong coupling in narrow-row operation. Studies that explicitly co-design the row representation, the local planner (e.g., DWA-like obstacle-aware planning), and the tracker (e.g., MPC, preview-based controllers, or adaptive FC) tend to report improved stability in cluttered or low-light conditions.

Importantly, the detailed records in Appendix Tables A4 and A5 also reveal fragmented evaluation practices: many works report detection accuracy, segmentation accuracy, or tracking error, but omit comparable measures of safety margin, clearance violations, failure recovery, or end-to-end navigation stability. This limitation is also observable in Table 2 and Fig. 7, where the reported KPIs are not consistent across the perception, planning, and control layers. Detailed study-level records for crop-row-guided path extraction and navigation-line generation are provided in Appendix Tables A4 and A5, respectively, while their cross-scenario algorithm and sensor patterns are synthesised in Figs. 8 and 9.

4.3.3. Local planning for row changes and headland manoeuvres

Row changes and headland manoeuvres govern the transition between within-row guidance and field-scale coverage, and they are often the most failure-prone phase of row-based autonomy. Turn trajectories must satisfy vehicle kinematics, implement constraints, and safety margins near boundaries, while minimising non-productive travel, overlap, and soil compaction. While substantial progress has been made in within-row perception and tracking, the coupling among row-end detection, headland free-space estimation, and turn-trajectory generation has been addressed inconsistently across the literature. With static information only, row changes are typically executed using template-based manoeuvres (e.g., circular, Ω-shaped, or S-shaped turns) parameterised by row spacing, machine dimensions, and minimum turning radius. Such templates are widely implemented in commercial guidance systems and can be effective in regular fields with low obstacle density. However, their performance degrades when field boundaries are irregular, rows are missing, or temporary obstacles appear—conditions that motivate real-time or hybrid adjustments.

Real-time and hybrid-single systems increasingly use sensor feedback to adapt row-end detection and turning trajectories. Representative studies incorporate perception-derived landmarks (e.g., tree trunks, row ends, or headland banks) and optimise turn curves using least-squares fitting, Bézier-based smoothing, or nonlinear optimisation. For example, Zhang et al. (2019) used LiDAR and inertial sensing with optimisation-based localisation to support row-change manoeuvres under canopy occlusion. Luo et al. (2022) employed stereo vision and hue–saturation–value (HSV)-based region-of-interest extraction to detect row ends for harvester turning. Gao et al. (2023) combined corner detection with Bézier-curve fitting to improve headland trajectories in irregular fields. More recent work integrates segmentation-based perception with turn planning as a coupled pipeline; Wu et al. (2025), for instance, reported high segmentation accuracy for headland-bank detection and centimetre-scale deviation in row-change execution, illustrating the potential of hybrid-single designs to adapt turning behaviour without recomputing the entire global coverage plan.

Beyond perception, several studies emphasise that headland

performance is ultimately limited by tracking robustness and actuation constraints. Controllers such as feedforward–PID, SMC, and MPC variants are therefore frequently reported to reduce overshoot and improve continuity during turns. Nevertheless, multi-vehicle interaction at headlands remains underexplored in the reviewed literature: even when multiple machines operate in the same field, headland conflict is often mitigated by manual scheduling or simple temporal offsets rather than real-time negotiation. This limitation motivates the subsequent multi-vehicle discussion in Section 5 and the cross-cutting analysis of evaluation protocols in Section 6.

Detailed records of recent studies on row-change path planning and tracking are provided in Appendix Table A6. These studies mainly combine short-range perception, vehicle-specific kinematic constraints, and feedback control, and are reflected in Figs. 8 and 9 through the tracking-control, vision/image-processing, LiDAR, GNSS/RTK, and stereo-vision categories.

4.4. Hybrid-information-based path planning and control for single-vehicle systems

In hybrid-information-based navigation, static priors (e.g., field boundaries, row geometry, headland layouts, or infrastructure records) are fused with real-time onboard sensing to improve navigation robustness in dynamic and partially known environments. Compared with purely static or purely real-time pipelines, hybrid-single systems explicitly separate (i) long-horizon structure encoded in priors (where the vehicle should generally go) and (ii) short-horizon variability captured online (the vehicle's current safe operational capabilities). This design is particularly relevant in orchards, greenhouses, and mixed-use farm corridors, where canopy occlusion, temporary obstacles, and seasonal geometry changes make either source alone insufficient.

From a systems perspective, hybrid-single autonomy can be viewed as a closed-loop stack with three tightly coupled layers: (i) online mapping and localisation for consistent ego-motion and workspace representation, commonly implemented via SLAM (Section 4.4.1); (ii) local replanning that adapts the reference trajectory using the updated map while respecting vehicle kinematics and safety constraints; and (iii) fusion-aware path tracking and control that compensates for slip, uneven terrain, and sensing uncertainty during execution (Section 4.4.2).

4.4.1. Slam-based navigation and local replanning

SLAM has emerged as a core enabler for hybrid-single navigation, especially where static maps are incomplete, outdated, or rapidly violated by operational changes. In agriculture, this need is amplified by seasonality (canopy growth, row closure, and varying understory), intermittent GNSS visibility, and temporary structures in protected environments. By jointly estimating the vehicle pose and building an online map, SLAM provides the planner with a continuously updated representation that bridges the gap between “as-designed” priors and “as-is” field conditions.

Recent studies illustrate that SLAM is used less as a standalone algorithm and more as a map-generation backbone that feeds conventional graph- or sampling-based planners. For example, SLAM-supported navigation has been reported for Ackermann-steered vehicles and orchard platforms, enabling stable local trajectory generation and improved motion execution under occlusion (Carpio et al., 2020; Xia et al., 2023). In smart-farm and greenhouse scenarios, online grid maps derived from SLAM or LiDAR-based mapping are frequently paired with classical search or local-avoidance planners, such as Dijkstra, A* search, rapidly exploring random tree star (RRT*), and the DWA, to balance path length, clearance, and crop protection (Pak et al., 2022; Liu et al., 2024; Guo et al., 2024a; Wang et al., 2024d).

A notable engineering trend is the move toward LiDAR–inertial SLAM and tightly coupled sensor fusion to maintain localisation continuity under GNSS degradation. For orchard operations, 3D LiDAR SLAM with point-cloud registration has been adopted to improve map consistency and row-level navigation reliability (Jiang et al., 2024). In confined or visually repetitive environments (e.g., stacked-cage farming houses), SLAM performance becomes highly sensitive to computational budgets and feature degeneracy, highlighting a persistent trade-off between mapping fidelity and real-time feasibility (Jiang et al., 2025a).

Despite these advances, three limitations remain recurrent in SLAM-enabled agricultural navigation. First, long-horizon drift and map ageing are exacerbated by repetitive textures and seasonal geometry changes, which can weaken loop closure and place recognition. Second, computational load increases sharply with high-resolution 3D mapping, limiting deployment on embedded controllers. Third, system integration (map management, replanning triggers, and safety fallbacks) is often under-specified, even when component-level performance is strong. These issues motivate more explicit architectural reporting and benchmark protocols that jointly evaluate mapping, replanning, and execution stability.

Representative studies on SLAM- and online-mapping-enabled planning are listed in Appendix Table A7. In the main text, these studies are incorporated into the SLAM/localisation category in Fig. 8, which shows their concentration in orchards, greenhouses, and mixed or facility-like scenarios where GNSS degradation, repetitive geometry, and confined manoeuvring make online mapping especially valuable.

4.4.2. Path tracking and navigation control using multi-source information fusion

Path tracking and control close the loop between the planned trajectory and the realised vehicle motion. In agricultural settings, execution disturbances are dominated by wheel slip, payload variation, uneven terrain, and intermittent sensing (particularly under canopy or indoors). Hybrid-single systems therefore increasingly rely on multi-source fusion to stabilise state estimation and to provide control-relevant signals (e.g., lateral error, heading error, slip proxies, or obstacle proximity) under partial observability.

In practice, most reported systems combine GNSS (often RTK-GNSS) with an IMU, wheel/steering encoders, and sometimes LiDAR or vision cues, using Kalman-filter-family estimators to maintain consistent pose and velocity estimates. On top of this estimation layer, two controller families dominate. The first comprises geometric tracking controllers, such as Pure Pursuit and Stanley-type controllers, which are attractive for low-speed operation due to their simplicity and predictable behaviour, but are sensitive to slip and curvature transitions. The second comprises model-based controllers, notably nonlinear MPC and SMC, which incorporate vehicle dynamics and constraints and tend to deliver better robustness under slip and terrain variation, at the cost of higher computational complexity and tuning effort (Ding et al., 2022; Soitinaho and Oksanen, 2023).

Learning-based and adaptive methods are also emerging, typically targeting regime changes that are difficult to model explicitly (e.g., sloping terrain and variable traction). For instance, data-driven modelling and reinforcement-learning-oriented strategies have been explored to improve performance under complex terrain and uncertain dynamics (Badgujar et al., 2023). However, their practical deployment is still constrained by data requirements, domain transfer, and the need for safety guarantees.

Overall, the literature suggests that the primary limitations on hybrid-single control performance stem not from the availability of individual algorithms but from system coupling issues: estimator latency, replanning frequency, controller bandwidth, and the discrepancy between offline priors and online geometry changes. This reinforces the need for reporting end-to-end timing budgets (perception–estimation–planning–control), as well as standardised evaluations that jointly quantify tracking stability, safety margins, and recovery behaviour under GNSS outages and occlusion.

Representative studies on navigation path tracking and control are provided in Appendix Table A8. These records indicate that tracking performance is increasingly stabilised by multi-source information,

including RTK-GNSS, IMU/INS, LiDAR, RGB-D vision, and wheel or steering encoders. In the graphical summaries, these studies contribute to the tracking-control category in Fig. 8 and to the GNSS/RTK, IMU/INS, LiDAR, RGB-camera, and RGB-D/stereo categories in Fig. 9.

5. Multi-vehicle cooperative navigation and path planning

Autonomous navigation has moved from proof-of-concept prototypes to operational systems in agricultural robotics, with single-vehicle platforms already deployed for guidance, spraying, and transport. As farm size increases and field operations become more time-constrained, however, multi-vehicle cooperation is progressively treated as an enabling capability rather than an optional extension. Coordinated fleets can raise effective field capacity, reduce labour dependency, and support compound operations (e.g., cooperative harvesting or concurrent tillage–seeding) that are difficult to achieve with a single machine.

This section reviews multi-vehicle cooperative navigation and path planning with an emphasis on static-information-based approaches, where the environment is represented by GNSS surveys, satellite imagery, pre-acquired field boundaries, or infrastructural maps. Under this assumption, cooperative planning can be decomposed into three tightly coupled subproblems:

(i) field partitioning and task allocation,
(ii) route coordination with spatial–temporal conflict avoidance, and
(iii) fleet-level optimisation that remains scalable in computation and communication as the number of vehicles grows.

In addition, this section highlights how real-time and hybrid information is increasingly introduced to relax the limitations of purely static representations, enabling more adaptive and robust multi-vehicle solutions. The section is organised accordingly, starting from static-map-driven methods and then moving toward real-time and hybrid cooperation mechanisms.

5.1. Static-information-based cooperative path planning for multiple vehicles

In large-scale and time-critical agricultural operations, static-map-driven cooperation offers an attractive engineering baseline because it enables predictable coverage, deterministic coordination rules, and offline optimisation. When the environment is reliably represented by field boundaries, headlands, lane geometry, and known obstacles, cooperative planning primarily revolves around: (1) how to split work fairly and efficiently, (2) how to avoid conflicts at shared resources (e.g., headlands, narrow corridors, or loading/unloading points), and (3) how to keep computation and communication tractable as fleet size increases. This section reviews static-information-based approaches from these three perspectives.

5.1.1. Field partitioning and task allocation on static maps

Field partitioning and task allocation constitute the foundation of cooperative operation under static representations. The planning problem is commonly reformulated as a constrained multi-travelling-salesperson problem (mTSP) or vehicle-routing problem (VRP), where each machine receives a subset of tracks, regions, or waypoints subject to workload balance, distance limits, precedence constraints, or safety constraints.

Lu et al. (2024) formulated allocation as a node–workload-constrained mTSP and solved it using a graph neural network (GNN) with a reinforcement-learning attention policy, aiming to minimise fleet makespan while balancing workload. For operations with explicit temporal precedence, synchronous complete-coverage strategies have been reported for dual-tractor tillage–seeding systems: tasks are decomposed into short "strip + turn" units and scheduled using geometric conflict detection to enable parallel, collision-free passes (Soitinaho et al., 2024). Decentralised coordination has also been explored. Guevara et al. (2021b) discretised fields into harvesting zones and dynamically assigned trailers to tractors to balance load under productivity variability, while Jo and Son (2025) proposed conflict-based hierarchical tasking that imposes constraints only on lower-priority agents, thereby enabling coordination with limited or no explicit communication in orchard scenarios.

Collectively, these studies indicate that when accurate static maps are available, task allocation can be treated as a combinatorial optimisation problem whose solution quality depends on how workload balance, precedence, and collision constraints are encoded. A key limitation is that most formulations implicitly assume that field geometry and productivity remain stable during execution; this reduces robustness when crop growth, soil conditions, machine availability, or operation delays vary over time.

5.1.2. Formation control and coordinated coverage on static representations

Beyond assigning disjoint regions to individual vehicles, formation control and coordinated coverage regulate inter-vehicle spacing and coordinated motion patterns during simultaneous operation. Using geographic information system (GIS) data, Jani (2020) demonstrated that formation strategies that explicitly regulate spacing can improve fleet efficiency and reduce idle time in cooperative field operations.

A common direction is to embed formation behaviours into hierarchical planning pipelines that decouple global coordination from local motion control. Ryu (2020) proposed a layered two-dimensional planner that builds navigable graphs from segmented local grids and safe intersections, separating inter-region routing from intra-region search to improve computational efficiency on large static fields. Lei et al. (2022) integrated satellite-image-based mapping, deep-learning-based global path generation, UAV-assisted obstacle sensing, and on-board LiDAR local avoidance into a unified hierarchical architecture, showing that off-board perception can reduce on-board complexity while maintaining robustness to static obstacles. In sloping vineyards, Mammarella et al. (2022) used UAV-derived 3D point clouds to construct simplified crop–terrain models and supported decentralised aerial–ground coordination, demonstrating improved operational efficiency in complex topography.

These results suggest that formation control on static maps is most effective when paired with hierarchical planning structures that separate fleet-level coordination from vehicle-level control. Nevertheless, most reported systems rely on predefined formation templates or fixed interaction rules; adaptive formation reshaping in response to workload imbalance, local bottlenecks, or unexpected disturbances remains uncommon in static-only settings.

5.1.3. Task-specific optimisation for cooperative operations

A third research stream focuses on task-specific optimisation tailored to particular agricultural objectives. Utamima et al. (2025) developed swarm-intelligence heuristics for seeding operations and reported shorter operation time and reduced deadheading distance compared with classical genetic and ant-colony algorithms, supporting near-real-time fleet coordination. For multi-robot planting in irregular fields, Liu et al. (2025a) combined minimal convex sub-region decomposition with shortest-path connection via Dijkstra's algorithm and an enhanced Non-dominated Sorting Genetic Algorithm III (NSGA-III), improving area balance and reducing redundant coverage.

Related studies incorporate terrain-aware cost functions into spanning-tree refinements or multi-objective ant-colony optimisation, explicitly accounting for turning effort, energy consumption, and overlap avoidance; these strategies report reductions in travel distance and energy use on static maps (Baras and Dasygenis, 2023; Yang et al., 2025). Importantly, these task-oriented approaches demonstrate that cooperative performance improves when agronomic or operational

objectives, such as minimising soil compaction, synchronising seeding and fertilisation, and limiting headland congestion, are encoded directly into the optimisation criteria rather than treated as post-hoc constraints.

Overall, static-information-based cooperative planning is effective in structured or slowly changing environments and enables strong optimisation for coverage, distance, and energy when high-quality maps are available. Representative formulations, algorithms, sensors, platforms, information types, and scenarios for cooperative planning are provided in Appendix Table A9. These records are synthesised graphically in Fig. 11, which maps cooperative studies across information source, algorithm category, platform configuration, and application scenario. However, performance remains strongly dependent on the accuracy and timeliness of the static representation: meaningful changes in field conditions typically require recomputing both task allocation and routes, and large-scale multi-season validation across heterogeneous farms remains limited. These limitations motivate the real-time- and hybrid-information-based cooperative strategies reviewed in Section 5.2, where static priors are complemented by on-board perception and inter-vehicle communication to cope with dynamic and partially known agricultural environments.

5.2. *Multi-vehicle cooperative path planning based on real-time and hybrid information*

Compared with static-information-based cooperative planning, real-time and hybrid-information-driven multi-vehicle navigation explicitly incorporates online perception, inter-vehicle communication, and dynamic system states into fleet-level decision making. Building on the single-vehicle real-time and hybrid paradigms reviewed in Section 4 and the static-multi strategies discussed in Section 5.1, this section analyses how recent studies extend cooperative planning toward scenarios with time-varying workloads, dynamic obstacles, and partially known environments. As indicated by the quantitative analysis in Section 3.4 and Table 1, real-time-multi and hybrid-multi systems still represent only a small proportion of the literature, suggesting that robust, perception-driven fleet coordination remains an open research frontier.

5.2.1. *Graph- and schedule-based cooperative planning with real-time constraints*

A first group of studies models multi-vehicle cooperation as a graph- or schedule-optimisation problem in which static road networks, row structures, or headland polygons are augmented with real-time task and vehicle-state updates. In such systems, global maps are used to generate time-expanded graphs, while online information modifies node weights, edge costs, and/or task availability.

Typical applications include the coordination of combine harvesters with grain carts and the scheduling of spraying or transport operations. Cao et al. (2021) proposed a dynamic ant colony optimisation (ACO) framework that integrates static field maps with online updates of machine workload and task completion, thereby reducing both travel distance and harvesting cycle time. Their subsequent work introduced terrain constraints and time-window reasoning to perform safety-aware conflict detection and global scheduling under real-time conditions (Cao et al., 2023). Similar time-expanded formulations are used in cooperative logistics to jointly minimise travel distance, fuel consumption, and makespan under continuously updated task states.

Despite incorporating online information, most graph- and schedule-based approaches still rely heavily on pre-acquired maps. Real-time data are commonly used to adjust priorities or reschedule tasks, rather than to trigger fine-resolution geometric replanning. Consequently, these systems remain closer to the static-multi corner than to fully perception-driven real-time-multi coordination in the analytical framework.

5.2.2. *Convoy and leader–follower strategies with perception assist*

A second research stream focuses on convoy- and formation-based cooperative navigation, where multiple machines operate in a leader–follower configuration. In these systems, one vehicle follows a predefined global route, while the remaining machines maintain prescribed

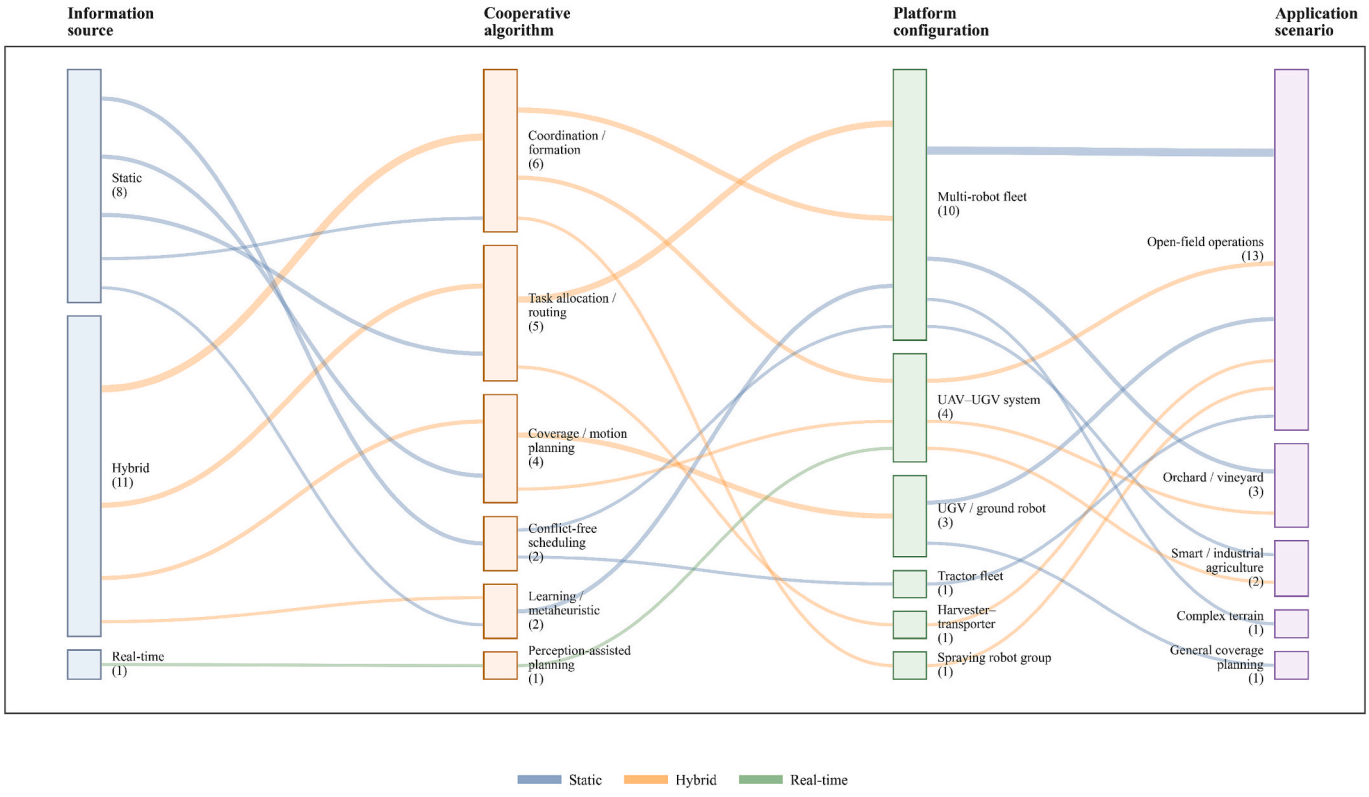


Fig. 11. Information-source and algorithmic structure of cooperative multi-vehicle path-planning studies.

offsets or formation geometry using real-time localisation and/or perception.

Lytridis et al. (2022) demonstrated modular cooperative strategies for heterogeneous vineyard machinery, using GNSS positioning and on-board sensing to regulate inter-vehicle distances and maintain collision-free operation. Qin et al. (2022) proposed a master–slave control architecture for a spraying robot group, integrating geometric guidance (pure pursuit) with feedback control to achieve stable convoy navigation on pre-constructed maps. In these applications, real-time GNSS, LiDAR, or vision measurements are essential to compensate for localisation drift, wheel slip, and terrain irregularities, enabling the convoy to remain cohesive in cluttered field environments.

Although perception-assisted leader–follower schemes move cooperative systems toward the real-time-multi category, global routes are typically fixed in advance, and interactions at headlands, intersections, or bottlenecks are often handled by heuristic right-of-way rules rather than explicit multi-agent optimisation.

5.2.3. Hybrid-multi architectures under limited communication and computation

Hybrid-multi systems aim to combine the global efficiency of static planning with the robustness of real-time perception. In a common pattern, static complete-coverage or graph-based planners are used to partition the field into subregions, while local navigation and collision avoidance rely on on-board sensors and inter-vehicle communication.

Air–ground collaboration is frequently used to reduce on-board perception burden. Liu et al. (2019b) coordinated an uncrewed aerial vehicle (UAV) with an uncrewed surface vehicle in rice paddies, using vision-based localisation (visual markers and perspective transformation) to enhance robustness. Mansur et al. (2025) employed UAV-derived row priors to accelerate cooperative coverage in row crops. Ren et al. (2024) discussed hierarchical UAV–UGV–UAV architectures that separate global mission planning, mid-level traffic management, and low-level feedback control, enabling static and real-time information to be fused at different layers.

In more challenging terrain, Yang et al. (2025) constructed a two-and-a-half-dimensional (2.5D) grid representation of unstructured fields and applied multi-objective ACO for global planning, while a genetic-algorithm-optimised DWA handled local obstacle avoidance and priority-based conflict resolution. Learning-based approaches further attempt to address communication and computation limits: Nivetha et al. (2025) introduced a self-supervised GNN to predict feasible paths from semantic aerial data and integrated it with online planners to improve stability under limited bandwidth.

Overall, hybrid-multi architectures appear to be the most promising direction for scalable cooperative navigation. However, consistent with Table 1, shared perception, distributed SLAM, communication-aware task allocation, and joint optimisation of logistics and navigation are still rarely implemented in practice. Bridging this gap requires planners that treat real-time perception and inter-vehicle information exchange as first-class fleet-level inputs, enabling routes, task assignments, and interaction policies to adapt online to changing crops, terrain, and operational conditions. Representative cooperative systems and algorithms are provided in Appendix Table A9 and further synthesised in Fig. 11. This graphical summary highlights the structural imbalance of cooperative navigation more clearly than a study-by-study table, showing that hybrid-multi systems remain promising but still limited in terms of shared perception, distributed mapping, and real-time fleet-level replanning.

6. Discussion

This section synthesises the findings of Sections 3–5 from a system-level perspective and identifies the main technological trajectories and research gaps in autonomous navigation for agricultural ground machinery. Rather than reiterating individual algorithms, the discussion is organised around three evolution lines—information sources, system architectures, and operation modes—and explicitly maps them back to the two-dimensional framework of information type × operation mode introduced in Section 2.6. Based on this synthesis, we outline the fundamental constraints that currently limit field-scale deployment and propose future research directions.

6.1. Evolution of information sources: From static priors to hybrid perception

The first and most evident evolution line concerns how environmental information is acquired and exploited for path planning. Early systems relied predominantly on static priors (e.g., field boundaries, surveyed rows, and pre-generated coverage tracks) and therefore concentrated in the static-single quadrant of the framework. As shown in Section 3.4, static-single approaches still account for a considerable share of the literature, particularly in open-field complete coverage and headland planning tasks.

Over the past five years, the dominant trend has shifted toward real-time perception, with RTK-GNSS, cameras, and LiDAR forming the backbone of single-vehicle navigation stacks. The statistical overview in Section 3 indicates that real-time-single systems now represent the largest category among the 210 analysed studies, reflecting rapid maturation in crop-row detection, obstacle perception, and localisation. Typical configurations combine RTK-GNSS with vision- or LiDAR-based perception to stabilise heading and position under partial signal occlusion, especially in orchards and greenhouses. This pattern is visible in the sensor–scenario distribution in Fig. 9 and is further supported by the detailed SLAM and path-tracking records in Appendix Tables A7 and A8.

More recently, hybrid-information-based navigation has emerged, in which static priors and real-time perception are tightly coupled. These systems anchor online perception to prior maps and enable local replanning when unexpected obstacles, row deviations, or terrain changes are encountered. Although hybrid-single studies are increasing, Section 3.4 shows that they still represent a minor fraction of the literature, and genuinely hybrid-multi systems remain rare. This imbalance suggests that, while the community increasingly recognises the need for hybrid information, its systematic exploitation at the fleet level—via shared maps, shared uncertainty, and shared update policies—remains largely unexplored.

The bibliometric results further support this information-source evolution. As shown in the VOSviewer keyword co-occurrence network and density map (Figs. 3 and 4), keywords related to perception, sensor fusion, SLAM, path planning, localisation, and control form prominent high-density clusters. This pattern is consistent with the transition identified by the proposed two-dimensional framework, namely the shift from static-information-based planning toward real-time perception and hybrid-information navigation. In contrast, keywords associated with multi-robot cooperation, fleet coordination, task allocation, and cooperative planning are less dominant in the bibliometric maps. This weaker representation corresponds to the quantitative imbalance shown in Table 1, where most real-time and hybrid studies remain concentrated in single-vehicle systems, whereas real-time-multi and hybrid-multi categories are still underrepresented. Therefore, Figs. 3 and 4 do not merely provide an independent keyword overview; rather, they cross-validate the structural evolution and research gaps revealed by the information-source × operation-mode framework.

Overall, the information-source trajectory can be summarised as a shift from static priors to real-time perception and then to hybrid coupling. This trajectory is supported not only by the coded study distribution in Section 3.4, but also by the keyword co-occurrence patterns in Figs. 3 and 4. However, this transition is currently concentrated in single-vehicle settings, leaving a clear gap in hybrid-multi navigation.

6.2. System architectures: From modular pipelines to tightly coupled fusion

A second evolution line concerns the architectural organisation of navigation systems. Traditional agricultural navigation stacks are typically modular, with loosely coupled components for localisation, perception, planning, and control. This design is interpretable and maintainable, but it tends to propagate errors across modules: localisation drift degrades path tracking; perception dropouts corrupt the reference path; and planning assumptions become invalid when the environment changes. In practice, this manifests as a fragile chain from perception output stability to trajectory trackability and ultimately to control robustness, particularly under canopy occlusion, illumination transitions, and terrain-induced slip.

Recent studies increasingly advocate tightly coupled multi-source fusion architectures, where localisation, mapping, and planning are jointly optimised. SLAM-based navigation in orchards and greenhouses, detailed in Appendix Table A7 and synthesised in Fig. 8, exemplifies this trend: LiDAR–inertial odometry and visual–inertial odometry provide more stable pose estimates under occlusion and support local replanning on updated maps. Similarly, multi-sensor fusion in row navigation—combining cameras, LiDAR, GNSS/RTK, IMU/INS, and depth sensing—improves robustness across illumination changes and crop growth stages. This pattern is supported by the detailed records in Appendix Tables A4 and A6 and by the sensor–scenario summary in Fig. 9. Importantly, tightly coupled designs also make it possible to implement a more reliable operational loop of drift → relocalisation → replanning, which is difficult to guarantee in loosely coupled pipelines.

Nevertheless, the mapping in Section 3 indicates that such tightly coupled architectures are still predominantly deployed in real-time-single and hybrid-single systems. Distributed or cooperative SLAM, shared perception maps, and fleet-level fusion mechanisms appear only sporadically in the literature. Consequently, most multi-vehicle systems still operate on partitioned static maps rather than dynamically shared environment representations, fundamentally limiting adaptability in complex and changing farm environments.

This architectural gap highlights a key research frontier: designing scalable, communication-aware fusion frameworks that propagate perception updates across vehicles in real time without overwhelming on-board computation or rural-network bandwidth.

6.3. Operation modes: From single-vehicle autonomy to cooperative fleets

The third evolution line concerns the transition from isolated single-vehicle autonomy toward coordinated multi-vehicle operation. Section 3.4 shows that the overwhelming majority of published systems still fall into the static-single and real-time-single quadrants, as quantified in Fig. 6 and Table 1, whereas real-time-multi and hybrid-multi systems account for only a small fraction of the reviewed literature.

Single-vehicle navigation has reached a comparatively mature stage. In structured environments such as open fields and orchards, centimetre-level tracking or localisation accuracy is frequently reported when RTK-GNSS is combined with vision- or LiDAR-based perception and geometric or model-based controllers, as summarised in the representative KPI synthesis in Table 2 and visualised in Fig. 7. However, extending these capabilities to fleets of heterogeneous machines introduces challenges that cannot be addressed by simply replicating single-vehicle stacks. Beyond local path tracking, multi-vehicle operation entails global coordination, conflict resolution at headlands and farm roads, workload balancing, and the synchronisation of machines that vary in kinematics, working widths, and operational constraints. Current multi-vehicle systems remain largely dependent on static field partitioning or pre-defined schedules (Section 5.1). Even when real-time perception is incorporated, online data are often used primarily for local obstacle avoidance, while task allocation and route planning remain offline or only weakly adaptive. As a result, genuine real-time-multi and hybrid-multi navigation—where fleets dynamically adapt task allocation and routes based on shared perception—remains rare. This gap between single-vehicle maturity and fleet-level autonomy represents a fundamental bottleneck in scaling from experimental platforms to farm-wide deployment.

The graphical synthesis in Fig. 11, derived from the cooperative-planning records in Appendix Table A9, further confirms that multi-vehicle navigation remains concentrated in static and hybrid information settings. Most cooperative studies focus on task allocation, conflict-free scheduling, formation or coordination control, and coverage-oriented fleet planning, whereas real-time cooperative navigation based on shared online perception remains rare. This pattern is consistent with the imbalance identified by the information-source × operation-mode framework in Table 1.

6.4. Cross-cutting research gaps

Synthesising the three evolution lines reveals four major research gaps that cut across information sources, architectures, and operation modes.

(i) Limited exploitation of hybrid information at the fleet level.

Hybrid navigation is increasingly common in single-vehicle systems, but its extension to fleets remains nascent. Shared perception, cooperative SLAM, and map-level fusion are rarely implemented, preventing coherent fleet responses to dynamic obstacles, changing crop structures, and degraded satellite positioning conditions.

(ii) Insufficient integration of navigation with agronomic objectives.

Most planners optimise geometric or kinematic criteria (e.g., path length, smoothness, energy). Agronomic outcomes—soil compaction, controlled-traffic constraints, crop damage risk, and long-term field sustainability—are seldom modelled explicitly, leaving navigation decisions weakly coupled to production objectives.

(iii) Weak evaluation practices and lack of benchmarking.

Validation is typically limited to single fields, short campaigns, or simulations, with limited emphasis on cross-season, cross-crop, and cross-farm generalisation. Public datasets and standardised benchmarks for agricultural navigation—especially in orchards, greenhouses, and multi-vehicle settings—remain scarce, constraining fair comparison and reproducibility.

(iv) Computational and communication bottlenecks.

Advanced fusion back ends, SLAM pipelines, and learning-based planners are computationally demanding, yet many agricultural robots operate under cost- and power-constrained hardware. Meanwhile, cooperative fleets require reliable low-latency communication that is often unavailable in rural environments. The trade-offs among accuracy, latency, bandwidth, and energy consumption remain insufficiently characterised.

A further cross-cutting gap concerns the heterogeneity of KPI reporting and the lack of benchmarkable evaluation protocols. The representative KPI synthesis in Table 2 and Fig. 7 shows that many studies report centimetre-level lateral deviation, localisation accuracy, or path-extraction performance, but these values are difficult to compare directly because they were obtained under different platform scales, vehicle speeds, crop structures, terrain conditions, sensor configurations, and validation settings. For example, some studies report vehicle-level tracking errors, whereas others report upstream perception or path-extraction indicators, such as navigation-line detection accuracy, angular fitting error, processing latency, or frame rate. These indicators

are all relevant to autonomous navigation performance, but they do not measure the same layer of the perception–planning–control pipeline. The complete study-level records supporting these graphical and KPI summaries are provided in Appendix Tables A1-A9.

Therefore, Table 2 and Fig. 7 should be interpreted as a representative KPI synthesis rather than as a controlled benchmark or *meta*-analysis. The comparison nevertheless reveals an important methodological gap: current studies rarely report a common set of metrics that jointly covers lateral deviation, heading error, localisation accuracy, computation time, safety margin, coverage quality, and task efficiency. In addition, real-time performance is reported mainly in perception-oriented studies, whereas global planning and cooperative planning studies more often report task-level indicators such as coverage rate, path length, energy consumption, operation cycle, or completion time. This inconsistency limits cross-study comparability and makes it difficult to identify whether performance gains originate from better sensing, improved planning, stronger control, or more favourable test conditions.

6.5. Implications for future research and deployment

Addressing these gaps requires advances in both algorithm design and system-level engineering.

First, future navigation systems should treat hybrid information as a fleet-level first-class input rather than an auxiliary feature of single-vehicle autonomy. This entails cooperative SLAM, map sharing, and perception-fusion mechanisms designed for agricultural constraints such as sparse connectivity, heterogeneous machinery, and long-duration missions.

Second, agronomic objectives should be integrated into navigation optimisation, for example by embedding soil-compaction models, crop-damage risk, and trafficability maps into cost functions. This would move navigation beyond geometric efficiency toward sustainable, crop-aware decision-making.

Third, the community should invest in benchmark datasets and evaluation protocols spanning multiple crops, seasons, and farm types, with explicit inclusion of multi-vehicle scenarios. Without such benchmarks, robustness and transferability will remain difficult to assess.

Finally, future work must confront the engineering realities of computation and communication. Lightweight fusion pipelines, edge–cloud collaboration, and communication-aware planning will be essential to make cooperative navigation viable on real farms.

In summary, while single-vehicle autonomous navigation in agriculture is approaching technical maturity, the transition to large-scale cooperative fleets that exploit hybrid information remains at an early stage. Bridging this gap represents both the central challenge and the principal opportunity for the next generation of agricultural navigation systems.

In addition to benchmark datasets, a more standardised reporting protocol is needed to improve the comparability and reproducibility of agricultural navigation studies. At minimum, representative studies should report the operating scenario, crop type, platform size, vehicle speed, sensor configuration, validation setting, lateral deviation, heading error, localisation error, processing latency or FPS, and task-level indicators such as coverage rate, path length, operation time, and energy consumption. Perception-driven navigation necessitates reporting both upstream perception accuracy and downstream tracking error, since high detection accuracy does not necessarily ensure stable vehicle-level performance. For global or cooperative planning, route efficiency should be reported together with execution feasibility, safety constraints, and coordination cost. Such a reporting protocol would make future KPI comparisons more transparent and would support the development of shared benchmark datasets and scenario-specific evaluation standards for agricultural ground machinery.

7. Conclusions

This review systematically mapped recent research on autonomous navigation and path planning for agricultural ground machinery published between 2019 and 2025. Following a PRISMA-ScR workflow, 210 studies were included and coded using a two-dimensional analytical framework that classifies methods by (i) the dominant type of environmental information (static, real-time, or hybrid) and (ii) the operation mode (single-vehicle vs. multi-vehicle). Descriptive statistics (Fig. 5, Fig. 6, Table 1), together with scenario-driven synthesis across open fields, orchards/vineyards, and protected environments, were used to characterise the technical landscape, expose structural imbalances, and extract transferable design patterns. In addition, the representative KPI synthesis and graphical summaries in Table 2 and Figs. 7–11 further reveal performance heterogeneity, algorithm–scenario imbalance, sensor-configuration differences, and the limited maturity of real-time cooperative navigation.

Overall, the literature indicates that open-field single-vehicle navigation—particularly real-time systems—is approaching engineering maturity. RTK-GNSS-based auto-steer solutions, often complemented by camera or LiDAR perception, can achieve centimetre-level tracking accuracy and high operational efficiency for many arable tasks. Static-information complete-coverage planning has also advanced substantially, supporting irregular field geometries and multi-objective optimisation, while local real-time planners provide robust obstacle avoidance and row following under moderately structured conditions. In orchards, greenhouses, and mixed farm environments, diverse perception-and-control pipelines—commonly built on LiDAR/vision with SLAM—have demonstrated feasibility, and commercial systems already exist for several high-value crops and protected cultivation settings.

At the same time, the quantitative evidence reveals a pronounced imbalance across the six conceptual categories defined by the proposed framework. Whereas the majority of real-time and hybrid methods are still confined to single-vehicle systems, multi-vehicle cooperation remains dominated by static priors and is validated predominantly in regular, open-field scenarios. Only a small number of studies integrate fleet-level coordination with rich real-time or hybrid perception in orchards, complex terrain, or protected cultivation. Moreover, many contributions are evaluated in a single site and season using non-standard performance indicators and heterogeneous machinery configurations, which limits comparability and weakens the evidence base for robustness under long-term and real-world variability.

Synthesising the comparative findings in Sections 4–6, three scenario-dependent trajectories emerge that can serve as concrete design guidance. For open fields, research is evolving from purely geometric static planning toward hybrid strategies that fuse pre-acquired maps with onboard perception and explicitly model headlands, turning costs, and logistics constraints. For orchards and vineyards, the centre of gravity is shifting toward safe navigation in narrow, cluttered alleys under degraded GNSS, where 3D perception, SLAM, and terrain-aware control are essential to manage stability and operational risk. For greenhouses and other facility environments, navigation is dominated by short-range docking, GNSS-denied localisation, and human–robot coexistence, favouring infrastructure-assisted anchoring (e.g., UWB), marker-based methods, visual SLAM, and compact platforms with tight coupling between perception, planning, and actuation. Combined with the discussion of machinery characteristics and production scales in Section 6, these trajectories indicate that future systems should be co-designed around information type, operation mode, platform mechanics, and farm scale—rather than treating these dimensions as post hoc deployment details.

Looking forward, the synthesis supports five priorities that can serve as a practical roadmap for the field:

(i) From isolated autonomy to cooperative fleets. The next capability frontier is not merely improving single-vehicle accuracy, but scaling to cooperative fleets that combine static farm priors, shared real-time perception, and distributed decision-making. This requires architectures for multi-vehicle coverage, coordinated logistics (e.g., harvester–cart interaction), and interaction management at headlands, intersections, and bottlenecks under explicit constraints on bandwidth, latency, and safety supervision. Progress will depend on hierarchical task allocation, conflict-aware coordination policies, and degraded-communication strategies that preserve safe behaviour when connectivity is intermittent.

(ii) Hybrid-information planning as the default across scenarios. Hybrid strategies that couple long-horizon priors with dense local perception remain under-explored outside open-field settings, particularly in orchards, mountainous terrain, and protected cultivation. A critical direction is to design planners and controllers that systematically exploit high-resolution priors (e.g., UAV/LiDAR maps, crop-row semantics, terrain models) while remaining robust to map–reality discrepancies. This calls for quality-aware fusion, explicit modelling of map uncertainty, and phase-aware switching among static, real-time, and hybrid modes (e.g., transit vs. operation vs. headland manoeuvres), supported by integrity monitoring rather than fixed fusion weights.

(iii) Standardised evaluation and long-term, multi-site validation. To move beyond proof-of-concept demonstrations, methods should be evaluated using harmonised KPIs, because the representative KPI synthesis in Table 2 and Fig. 7 shows that current performance indicators remain highly heterogeneous across scenarios, platforms, sensors, and validation settings. Future evaluation should jointly quantify coverage quality, tracking error, localisation accuracy, computation time or FPS, safety margins, energy efficiency, and agronomic impact, such as soil-compaction and crop-damage proxies. Multi-site, multi-season, and cross-platform trials are essential to establish robustness across soils, crops, canopy conditions, and weather regimes. Community progress would be accelerated by shared benchmark scenarios, reporting templates for sensors and machinery, and reproducible baselines that make results cumulative rather than anecdotal.

(iv) Deeper integration with farm management, machinery design, and safety requirements. Many planning methods are still developed in isolation from whole-farm workflows, labour organisation, and compliance constraints. Future systems should interface directly with farm management information systems (FMIS), prescription maps, and work-order scheduling, while incorporating operational safety zones, human–robot interaction protocols, and traceability requirements. At the platform level, tighter co-design that links planning to machinery kinematics, implement dynamics, and maintenance constraints can improve both efficiency and equipment longevity, shifting optimisation targets from purely geometric paths to operationally meaningful trajectories.

(v) Learning-enabled navigation across the full stack, with safety-grounded fallbacks. The growing availability of multimodal data from machinery, field sensors, and remote sensing enables learning-based perception, prediction, and decision modules, but these components must be embedded within interpretable, safety-aware architectures. Promising directions include reinforcement learning and model-based learning for complex manoeuvres (row changes, docking, headland interactions), data-driven cost functions that better encode agronomic objectives (soil protection, crop-damage risk, fuel use), and adaptive controllers that use historical and online data to tune constraints and gains. A decisive need is principled uncertainty handling and certified fallbacks so that learning improves robustness without eroding safety and predictability.

In summary, autonomous navigation and path planning for agricultural ground machinery has advanced rapidly, with the strongest maturity observed for single-vehicle systems in structured open-field operations. By organising the literature within a unified information–operation-mode framework and integrating quantitative statistics with scenario- and scale-based synthesis, this review clarifies where technologies are close to deployment and where evidence remains insufficient. More importantly, it provides a forward-looking engineering agenda: make hybrid-information planning the default, elevate fleet cooperation to a first-class design target, standardise evaluation with multi-site and multi-season validation, integrate navigation tightly with farm management and machinery design, and adopt learning-enabled modules within safety-grounded architectures. Delivering on these priorities is essential for moving from isolated demonstrations toward robust, scalable, multi-vehicle, multi-scenario navigation systems that can underpin sustainable, large-scale digital agriculture.

CRediT authorship contribution statement

Xuepeng Xiao: Writing – original draft, Investigation. **Hongbo Yuan:** Writing – original draft, Methodology. **Yashuai Yang:** Writing – review & editing. **Man Cheng:** Writing – review & editing, Resources, Data curation, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

Detailed coding records and extracted performance records.

Table A1
Detailed summary of complete coverage path planning algorithms.

Authors (Year)	Data source / environment modelling	Optimisation criteria	Representative algorithm / method
Lu et al., 2020	Satellite imagery; polygonal approximation	coverage rate (CR); overlap; number of turns; efficiency	Polygon decomposition; coverage planning
Mazzia et al., 2021	Aerial imagery; fully convolutional network (FCN) waypoints; crop-row texture	waypoint error; row alignment; path feasibility	deep learning (DL); global planning
Wang et al., 2023b	LiDAR mapping; kiwifruit zoning	path length; picking efficiency; safety events	deep reinforcement learning (DRL); coverage planning
Martelloni et al., 2019	RTK-GNSS trajectories; lawn geometry	coverage efficiency; task time; overlap	Empirical assessment
Lin et al., 2025	UAV + multi-task YOLO	geolocation error; detection F1; route reliability	contour/shape rule extraction
Wang et al., 2024b	Pre-sowing priors; operation semantics	CR; distance ratio; smoothness	cell decomposition; arcs/splines
Xu et al., 2025	UAV tiles; row/plant spacing constraints	geometric consistency; path error; compute time	improved Dijkstra
Li et al., 2024e	Orchard point-cloud prior	tracking stability; lateral/heading error; speed	A* (global) + timed elastic band (TEB) (local)
Li, 2022b	2D grid	search time; turn minimisation; smoothness	bidirectional alternating A* search
Guo et al., 2024a	Greenhouse	planning time; smoothness; tracking error	weighted A* search + Bézier smoothing + Dynamic Window Approach (DWA)
Chatzisavvas et al., 2024	Generic 2D grid	smoothness; tracking stability	A* search + Bézier smoothing
Zhang et al., 2025d	Protected agriculture	compute time; CR	A* search + depth-first search (DFS)
Wang et al., 2024d	Greenhouse–orchard	steering complexity; pose error; safety	improved A* search + fuzzy Dynamic Window Approach (DWA)
Jiang et al., 2025b	Orchard	centreline deviation; curvature continuity	enhanced D* Lite + Hybrid A* search
Ait et al., 2023	Row crops; kinematics	completion time; kinematic feasibility	Travelling Salesman Problem (TSP) (global) + timed elastic band (TEB) (local)
Wang et al., 2025	E-map; convex plots	coverage completeness; time efficiency; path length	minimum row spacing nested method (MRS-NM) × inner-spiral hybrid
Zhang et al., 2024a	Orchard mower; 2D grid	compute cost; CR; smoothness	improved A* search + DFS
Zhang et al., 2025c	2D radar grid	path optimality; total turning; convergence speed	hybrid enhanced dung-beetle optimisation (HDBO) (metaheuristic)
Guo et al., 2025	Large-scale static grid	search speed; optimality; scalability	Super jump point search (S-JPS)
Zhang et al., 2024c	Field map (rows/roads/obstacles)	turning energy; planning time; feasibility	heuristic + dynamic programming
Chen et al., 2025a	Indoor	path length; number of turns; compute time	enhanced grey wolf optimiser (GWO) + improved Dynamic Window Approach (DWA)

Table A2
Detailed summary of multi-objective complete coverage path planning algorithms.

Authors (Year)	Application scenario	Optimisation criteria	Representative algorithm / method
Yang et al., 2024	Multi-region global traversal	collision-free; cost reduction	A* search + genetic algorithm (GA)
Kong et al., 2024	Orchard fertilisation	smoothness; trackability	improved A* search + Bézier curves
Xie et al., 2024	Horticultural pruning	path length; coverage; turns	hybrid genetic ant-colony + Boustrophedon + A* search
Mahmud et al., 2019	Greenhouse spraying	multi-objective performance	probabilistic roadmap (PRM) + non-dominated sorting genetic algorithm III (NSGA-III)
Zangina et al., 2021	Greenhouse	solution quality; runtime	improved NSGA-III (vehicle-routing formulation)
Zhang et al., 2022d	Agricultural machinery path planning	algorithm performance	HypE (multi-objective evolutionary algorithm)
Shangguan et al., 2021	Grain transport	effectiveness; practicality	artificial potential field (APF) + fuzzy-logic control (FLC)
Wei, 2021	Energy-constrained operations	energy awareness; energy mapping	deep neural network (DNN) + coverage planning
Wu et al., 2023	Irregular fields	reduced redundancy; fewer turns	improved genetic algorithm (GA)
Kutyrev et al., 2022	Horticultural systems	efficiency; autonomy; localisation accuracy	multi-index planning platform
Yuan, 2022	Orchard navigation	global path planning	image mosaicking + convolutional neural network (CNN) + LiDAR
Xie et al., 2025	Hilly orchards	energy-minimising multi-robot assignment	Golden Kepler optimisation algorithm (GKOA)
Hu et al., 2025	Horticulture	planning efficiency; smoothness; safety	LiDAR-based semantic SLAM + navigation stack

Table A3
Detailed coding records of real-time navigation information acquisition studies.

Authors (Year)	Sensor configuration	Platform	Representative algorithm / method	Application scenario
Han et al., 2019a	RTK-GNSS	Sprayer	—	Orchard
Blok et al., 2019	2D LiDAR; IMU; wheel encoders	Four-wheel differential steering	Particle filter (PF); Kalman filter (KF)	Orchard
Si et al., 2019	Micro-Electro-Mechanical Systems (MEMS); Global Positioning System (GPS)	Tractor	Unscented Kalman filter (UKF)	Field
Fue et al., 2020	RTK-GNSS; encoders; IMU	Four-wheel steering	modified Pure Pursuit; dual extended Kalman filter (EKF)	Cotton field
Lamb et al., 2022	Smartphone GNSS	Smartphone	Greedy + simulated annealing	Field
Cui et al., 2024	RTK (SPP)	Vehicle	EC-Kalman filter (KF)	Field/road
Wang et al., 2024a	GNSS; INS	Vehicle	OSS-Transformer (online semi-supervised learning)	Field/road
Winterhalter et al., 2021	Wheel encoders; IMU; RTK-GNSS; monocular camera	Four-wheel steering	GNSS + crop-row tracking fusion	Various vegetables
Xu et al., 2022b	RGB camera; RTK-GNSS; IMU	Four-wheel model	Line tracking + sensor fusion	Field
Li et al., 2021b	GNSS; IMU	Tractor	FA-FIR-Kalman filter (KF)	—
Suk et al., 2025	Monocular camera; RTK-GNSS; IMU; Hall sensors	UGV	Kalman filter (KF) fusion + preview tracking	Orchard
Tinchev et al., 2019	2D LiDAR	—	deep learning (DL)	Forest
Guevara et al., 2021a	3D sensor	—	3D point-cloud matching	Field
Thanh et al., 2022	2D LiDAR	Lawn mower	density-based spatial clustering of applications with noise (DBSCAN)	Field
Raikwar et al., 2022	Wheel and steering encoders	Four-wheel model	—	Orchard
de Lima et al., 2023	3D LiDAR (air/ground)	Outdoor robot	Monte Carlo localisation	Forest
Li et al., 2024d	RTK-GNSS; INS; LiDAR	UGV	dynamic GNSS/INS–LiDAR switching + proportional–integral–derivative (PID)	Orchard
Yao et al., 2024	LiDAR (Lightweight and Ground-Optimised LiDAR Odometry and Mapping (LeGO-LOAM)); multi-sensor	UGV	Regularised LeGO-LOAM + OEN planner (hysteresis)	Greenhouse
Fujinaga, 2025	LiDAR	Agricultural robot	Waypoint following + bed navigation	Strawberry high-bed farm
Zhang et al., 2024b	IR camera (850 nm); DRI tag; odometry	Mobile chassis	Marker-based localisation + U-shaped network (U-Net) yaw estimation	Poultry house (indoor)
Chen et al., 2024	UWB (6 anchors)	Feed-pushing robot	dynamic look-ahead Pure Pursuit + PID	Dairy barn
Yao et al., 2021	UWB	Mobile platform	fuzzy control (FC)	Greenhouse
Gao et al., 2022	IMU; GNSS	AMR	Kalman filter (KF) + autoencoder fusion	Road; grassland; field
Jing et al., 2023	GNSS; INS	Tractor	adaptive square-root cubature Kalman filter (ASRCKF)	Field
Xie et al., 2022	NB-IoT; BeiDou Navigation Satellite System (BDS)/ GPS	—	adaptive optimal zero suppression (AOZS) + dual-mode positioning	Campus
Xu and Li, 2022c	Dual GNSS antennas; RGB camera	Modular agricultural robot	Target tracking	Field

Table A4
Detailed summary of crop-row-guided path extraction methods.

Authors (Year)	Sensor	Algorithms	Scene
Higuti et al., 2019	2D LiDAR	Least-squares fitting	Corn and sorghum fields
Kanagasingham et al., 2020	GNSS; IMU; camera	Novel crop-row detection	Rice paddies
Ma et al., 2022	Monocular camera; GPS	Image segmentation; region search	Rice paddies
Gasparino et al., 2020	2D LiDAR; IMU	3D point-cloud modeling	Corn fields
Zhang et al., 2020	2D LiDAR	Euclidean clustering	Orchards
Jiang et al., 2022	Thermal imager	Faster R-CNN	Orchards
Calera et al., 2023	Monocular camera	Image processing	Orchards (under canopy)
Opiyo et al., 2021	RGB camera	K-means; Gabor filters	Orchards
Nehme et al., 2021	LiDAR	Hough transform (structure tracking)	Vineyards
Yun et al., 2021	Stereo camera; GPS	Pitch/roll compensation for stereo guidance	Fields
Velasquez et al., 2020	2D LiDAR	H∞ robust control (reactive navigation)	Corn fields
Gai et al., 2025	RGB camera	two-dimensional discrete Fourier transform / discrete-time Fourier transform (2D-DFT/ DTFT) + linear–quadratic–Gaussian (LQG) control	Row crops (varied lighting)
Zhang et al., 2022b	RTK–BDS; IMU	Loosely coupled Kalman filter	Orchards (sprayer)
Liu et al., 2025c	—	Split preview + variable-universe FC	Hilly terrain (monorail platform)
Aghi et al., 2020	RGB-D camera	Transfer learning	Vineyards (steep slopes)
Schmitz et al., 2022	Ultrasonic sensor	—	Corn fields
Le et al., 2020	2D LiDAR	Adaptive Monte Carlo localization (AMCL); waypoint-lite row following using RANSAC	Greenhouses / polytunnels

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Table A4 (*continued*)

Authors (Year)	Sensor	Algorithms	Scene
Donati et al., 2022	GPS; IMU; ultrasonic	Kalman-filter sensor fusion	Vineyards
Li and Wu, 2025	RGB camera	CGNet + self-attention semantic segmentation	Field road intersections (mountainous farmland)

Table A5
Detailed summary of navigation-line generation and path-tracking methods for crop-row-based navigation.

Authors (Year)	Sensor configuration	Representative algorithm / method	Application scenario
Chen et al., 2020a	RGB camera	Midpoint Hough transform	Greenhouse (tomato–cucumber)
Chen et al., 2020b	RGB camera	image segmentation + waypoint extraction	Corn fields
Gao et al., 2020	RGB-D / color-depth	Deep fusion segmentation (spray path)	Peach orchard
Emmi et al., 2021	2D LiDAR; RGB; odometer	Semantic localisation and key-pose estimation	Vegetable/vineyard
Liang et al., 2022	Camera	edge detection + Otsu thresholding	Cotton fields
Ruangurai et al., 2022	Monocular camera	principal component analysis (PCA) + Hough	Rice paddies
Zhang et al., 2022e	Stereo vision	edge detection for row boundary	Wheat/harvesting
Wang et al., 2023a	Monocular camera	Vegetation index + ridge-line segmentation	Lettuce fields
Zhou et al., 2025	Monocular camera	U-Net + convolutional block attention module (CBAM) (low-light path segmentation)	Trellis orchard
Kim et al., 2020	IP camera	convolutional neural network (CNN) (row/path detection)	Citrus orchards
Li et al., 2021a	CCD camera	Deep vision + improved Hough	Various crops
Kim et al., 2021	IP camera	convolutional neural network (CNN) (unharvested rice region guidance)	Rice paddies
Li et al., 2022c	Monocular camera	Faster U-Net (row count/segmentation)	Corn/Tomato/Cucumber/Wheat
Sun et al., 2022a	Monocular camera	SADNet (UAV image segmentation)	Orchards
Cao et al., 2022	Monocular camera	ENet (lightweight semantic segmentation)	Beet fields
Wei et al., 2022	Monocular camera	Row anchor selection classification method (RASCM)	Early-stage corn
Yu et al., 2023	Monocular camera	Semantic segmentation (robust to shadow/obstacles)	Fields
Zhong et al., 2024	Monocular camera	U-Net + CBAM (low-light path segmentation)	Trellis orchard
Li et al., 2024c	Monocular camera	VC-U-Net + least-squares centreline	Rapeseed fields
Zhu et al., 2024	Monocular camera	DeepLabv3+	Pitaya orchard
Cao et al., 2024	Monocular camera	YOLOv8-Trunk + least-squares dual reference lines	Orchard (trunk-based)
Guo et al., 2024b	Monocular camera	YOLOv8s-CornNet + least-squares row midline	Corn (spraying)
Li et al., 2022a	RGB camera	improved deep Q-network (I-DQN) (deep reinforcement learning (RL) controller)	—
Xu et al., 2022b	UWB; IMU	Variable preview / fuzzy control (FC)	Greenhouses
He et al., 2022	RTK-GNSS; IMU	model predictive control (MPC) path tracking	Rice paddies
Xu et al., 2022a	—	efficiency-oriented model predictive control (EfiMPC)	—
He et al., 2023	GNSS; IMU	particle swarm optimisation (PSO)–support vector machine (SVM) tracking enhancement	Rice paddies
Guyonneau et al., 2022	LiDAR	PEARL/Ruby line extraction	Row crops
Li et al., 2020	Binocular camera	improved RANSAC-based path detection	rice/wheat
Leanza et al., 2023	IMU; GPS	Adaptive Kalman + linear regression (heading)	Fields
Yao et al., 2025	LiDAR + visual–inertial odometry (VIO) + camera	Stable feature localisation + PPM-U-Net (obstacles)	Unstructured farms
Santos et al., 2020	IMU; odometry; GNSS; 2D LiDAR; camera	AgRobPP (A* search on multi-layer maps)	Vineyards (slope)
Cerrato et al., 2024	Low-cost sensors (grid map)	End-to-end pipeline (occupancy path generation)	Row crops (GNSS-degraded)
Zhang et al., 2025a	UAV imagery	Weak-supervised dual-mask edge seg. + variable path	Soybean/weed rows
Tao et al., 2025	Monocular camera	CDSC-DeepLabv3+ + RANSAC-based centreline fitting	Tea canopy (hilly)
Kuang et al., 2025	Onboard camera (MAV)	insect-inspired histogram of oriented gradients (HOG) pipeline (target-guided)	Plantation (tree rows)
Mwitta and Rains 2024	GPS + camera	GPS-based global planning + deep learning (DL) segmentation + Dynamic Window Approach (DWA) local planning	Cotton fields
Chen et al., 2025b	3D LiDAR + IMU	LIO-SAM map → 2D grid + multi-point centreline navigation	Orchard (rows and headland)
Gao et al., 2025	—	graph methods + bio-inspired optimisation + learning	Dynamic scenarios (formation/ navigation)

Table A6
Detailed summary of row-change path planning and tracking studies.

Authors (Year)	Sensor configuration	Representative algorithm / method	Platform	Application scenario
Zhang et al., 2019	Gyroscope, LiDAR	extended Kalman filter (EKF)	Creeper truck	Orchard
Luo et al., 2022	Binocular camera	dynamic region-of-interest (ROI) extraction based on HSV space	Harvester	Rice, canola, corn
Wu et al., 2025	Stereo camera	DeepLabv3+	picking robot	Safflower
Guevara et al., 2020	–-	Cape turning algorithm for N-type trailers	N-type trailer	Field
Huan et al., 2020	GNSS	Feedforward plus proportional–integral–derivative (PID) control	Tracked vehicle	Field
Zhou et al., 2021	Monocular camera	Hough matrix and random sample consensus (RANSAC)	Tracked robot	Orchard

(*continued on next page*)

Table A6 (*continued*)

Authors (Year)	Sensor configuration	Representative algorithm / method	Platform	Application scenario
Li et al., 2023	GPS, IMU	sliding-mode control (SMC)	Rice transplanter	Rice
Gao et al., 2023	Binocular camera, RTK-GNSS, IMU	Multi-order Bézier curve	Picking robot	Safflower

Table A7
Detailed summary of SLAM-based path planning studies for agricultural machinery.

Authors (Year)	Application scenario	Optimisation criteria	Representative algorithm / method
Carpio et al., 2020	Orchard	Smoothness, stability	planner + attitude control
Zhao et al., 2020	Farm	Completeness, load balance	Mesh-SLAM
Pak et al., 2022	Greenhouse	Path length, avoidance	Dijkstra, A* search, RRT*
Xia et al., 2023	Trellised orchard	Accuracy	Pure Pursuit
Feng et al., 2023	Greenhouse	localisation, robustness	RBPF-SLAM, Waypoint planning
Liu et al., 2024	Greenhouse (tomato)	Accuracy	A* search, Dynamic Window Approach (DWA), adaptive Monte Carlo localisation (AMCL)
Zhu et al., 2023	Orchard	Place recognition	Memory network
Zhu et al., 2025	Orchard	Robustness, avoidance	extended Kalman filter (EKF) fusion, ROS
Jiang et al., 2025a	Stacked cage	Accuracy, compute	Faster-LIO, VINS-Fusion
Koc and Vatandas, 2025	Farm logistics	Deviation, safety	extended Kalman filter (EKF), SLAM, A* search, Dynamic Window Approach (DWA)
Wang et al., 2024c	Orchard	Accuracy, safety	A* search, Dynamic Window Approach (DWA), Kalman filter (KF) fusion
Jiang et al., 2024	Orchard (spindle-shaped)	Accuracy, error	NDT–ICP, ROS
Liu et al., 2025b	Orchard	Speed, parameters	NPENet

Table A8
Detailed summary of navigation path-tracking algorithms.

Authors (Year)	Sensor configuration	Platform	Representative algorithm / method	Application scenario
Han et al., 2020	RTK-GNSS	Tracked agricultural vehicle	multi-sensor fusion	Field
Yin et al., 2020	RTK-GNSS	Agricultural vehicle	—	Field
Han et al., 2019b	RTK-GNSS; IMU	Tractor	Slip estimation	Rice paddy
Fernandez et al., 2019	GNSS; Hall sensor	Tractor	Observer-based optimal control	Field
Zhou et al., 2020	Monocular camera	Tractor	robust controller	Field
Liu et al., 2019a	GPS; IMU; Tacho-generator	Irrigation machine	updated Sage-Husa adaptive Kalman filter (KF)	Field
Wang et al., 2020	GPS; Gyro	Tractor	proportional–integral–derivative (PID) controller	—
Han et al., 2022	IMU; GNSS	Tracked vehicle	sensor fusion	Orchard
Ding et al., 2022	GPS	Tractor	Output-feedback sliding-mode control (SMC)	Field
Ji et al., 2022	—	Agricultural vehicle	new sliding-mode controller (SMC)	Field
Zhang et al., 2022a	RTK-GNSS; IMU; Ultrasonic	Tracked vehicle	Feedback control (pole placement)	Orchard
Zhang et al., 2022c	IMU; GNSS	Four-wheel-steering (4WS) agricultural machine	fuzzy control (FC)	Field
Zhang et al., 2021	—	Harvester	Adaptive neural network	Wheat and Rice
Yang et al., 2022a	GPS	Tractor	Pure Pursuit	Field
Liu et al., 2019c	2D LiDAR	Tractor	nonlinear model predictive control (NMPC)	—
Soitinaho and Oksanen, 2023	RTK-GNSS	Tractor	nonlinear model predictive control (NMPC)	Field
Zhao et al., 2023	RTK-GNSS; IMU; 2D LiDAR	Tractor	Minimum capture algorithm	Field
Chen et al., 2023	RTK-GNSS; 2D LiDAR	Tractor	recovery-machine kinematic model	Cotton
Sun et al., 2022b	Dual-antenna RTK	Combine harvester	PSO	Field
Lohan et al., 2022	GPS	Rice transplanter	—	Rice paddy
Badgujar et al., 2023	GPS; RTK	Tractor	machine learning algorithm	Field
Li et al., 2024d	RTK-GNSS; INS; LiDAR	Orchard robot	Dynamic GNSS/LiDAR switching + proportional–integral–derivative (PID) path tracking	Orchard
Li et al., 2024a	RGB-D	Field vehicle	ESANet RGB-D segmentation + slope-adaptive feature and LS fit	Irregular small fields (cape)
Mwitta and Rains 2024	GPS; Monocular camera	Ackermann mobile robot	GPS + Pure Pursuit / fully convolutional network (FCN) + Dynamic Window Approach (DWA) / integrated (global + local)	Cotton field
Gai et al., 2025	Monocular camera	Field robot	2D-DFT/DTFT row detection + LQG tracking	Row-crop field
Zhang et al., 2025b	GNSS; IMU	Tracked rice harvester (YH1180)	Multi-parameter optimised nonlinear PID	paddy fields
Lin et al., 2025	UAV imagery (offline); GNSS (deploy)	Weeding UGV	Multi-task YOLO (navigable-area and weed) → GNSS waypoints	Pineapple plantation

Table A9
Detailed summary of cooperative algorithms for multi-vehicle path planning.

Authors (Year)	Sensor configuration	Representative algorithm / method	Platform	Information type	Application scenario
Lu et al., 2024	GIS map	graph neural network (GNN) + reinforcement learning (RL) attention policy (mTSP allocation)	Robot fleet	Static	Farm operations
Jo and Son, 2025	—	Conflict-based hierarchical tasking (CBS-HT) (prioritised safe-interval planning)	Heterogeneous UGV fleet	Static	Orchard / sorting / warehouse
Soitinaho et al., 2024	—	Synchronous CCPP scheduling + conflict rules	Two autonomous tractors	Static	Field (tillage–seeding)
Jani, 2020	GIS map	Formation control (spacing regulation)	Agricultural machinery fleet	Static	Field operations
Guevara et al., 2021b	—	Decentralised trailer–tractor coordination	N-trailer vehicle fleet	Hybrid	Field (harvesting logistics)
Cao et al., 2021	GNSS / field map	improved ant colony optimisation (ACO) (dynamic task allocation)	Harvester + grain transporter	Hybrid	Grain fields
Cao et al., 2023	Topographic map / time window	Dijkstra + time-window conflict detection	Multi-machine fleet	Hybrid	Farmland (safety-aware scheduling)
Lytridis et al., 2022	GNSS; 2D LiDAR (simulation setting)	Modular cooperative navigation	Heterogeneous robots (simulation)	Hybrid	Vineyard
Qin et al., 2022	Map prior; on-board ranging (reported)	Pure Pursuit + master–slave coordination	Spraying robot group	Hybrid	Field spraying
Liu et al., 2019b	UAV vision; on-board camera	Vision-based localisation (UAV-assisted)	UAV + surface vehicle	Real-time	Rice paddy
Mansur et al., 2025	UAV RGB imaging	UAV row mapping + coverage support	UAV + ground machinery	Hybrid	Maize / soybean fields
Ren et al., 2024	Multimodal (UAV + UGV) sensors	Hierarchical cooperative framework (survey)	UAVs; UGVs	Hybrid	Industrial-scale agriculture
Yang et al., 2025	2.5D grid; IMU	multi-objective ant colony optimisation (ACO) + genetic algorithm (GA)-optimised Dynamic Window Approach (DWA)	Multiple ground machines	Hybrid	Slope / rocky / unknown terrain
Ryu, 2020	—	SIRRT* (skeletonisation-informed RRT*)	Mobile robots (generic)	Static	Large, complex farmland
Lei et al., 2022	Satellite imagery; UAV sensing; on-board sensors	LiDAR- and deep-learning-based coverage planning (hierarchical)	Ground robot / vehicle (reported)	Hybrid	Field environments
Mammarella et al., 2022	UAV-derived 3D model / imagery (reported)	Air–ground cooperative navigation (case study)	UAV + ground vehicle	Hybrid	Vineyard (sloped)
Baras and Dasygenis, 2023	—	Turn-reduction / energy-aware coverage planning	UGV (generic)	Static	Coverage planning (general)
Utamima et al., 2025	—	Lovebird algorithm (swarm heuristic)	Seeding UGV fleet	Static	Field (seeding)
Liu et al., 2025a	—	Improved NSGA-II + region decomposition	Multi-robot UGVs	Static	Planting in irregular fields
Nivetha et al., 2025	Semantic aerial dataset	self-supervised graph neural network (GNN) path prediction + online planner	Multi-agent system (reported)	Hybrid	Smart agriculture

Table A10
Extracted performance improvements of representative navigation and path-planning algorithms.

Navigation / planning task	Baseline method	Improved / proposed method	Reported performance improvement	Main improved indicator	Reference
Local path tracking	Conventional Pure Pursuit with fixed look-ahead distance	ISSA-based improved Pure Pursuit / optimal look-ahead strategy	Lateral deviation reduced by 51.25%; heading deviation reduced by 30.62%; navigation time shortened by 10.67%	Lateral deviation; heading deviation; navigation time	Wen et al. (2025)
Local path tracking	Conventional Pure Pursuit with fixed look-ahead distance	UWB-based dynamic look-ahead Pure Pursuit	Mean lateral deviation reduced by 42.83%; RMSE reduced by 42.90%	Mean lateral deviation; RMSE	Chen et al. (2024)
Local path tracking	conventional sliding-mode / Stanley-type tracking baseline	PSO-enhanced fuzzy Stanley controller	Maximum lateral error reduced from 0.32 m to 0.03 m; faster trajectory response was reported	Maximum lateral error; response behaviour	Sun et al. (2022b)
Global–local hybrid planning	Conventional A* search	improved A* search + fuzzy Dynamic Window Approach (DWA)	Distance deviation reduced by up to 31.8%; heading-angle deviation reduced by up to 28.6%; critical turning points reduced by 40%	Distance deviation; heading deviation; turning complexity	Guo et al. (2024a)
Global–local hybrid planning	Conventional D* Lite	Hybrid D* Lite + A* search	Mean centreline deviation reduced by 75.55%; number of dangerous nodes reduced by 92.86%	Centreline deviation; dangerous nodes	Jiang et al. (2025b)
Global–local hybrid planning	Conventional A* search + SVR model	Improved A* search + SVR motion model	Mean lateral deviation was 6.90 cm and improved by 29.57%; longitudinal deviation improved by 10.12%	Lateral deviation; longitudinal deviation	Liu et al. (2025c)
Global planning and time optimisation	Conventional A* search	Improved A* search + DFS	Computation time reduced by 30%; key nodes reduced by 50%; coverage rate increased by 7.9%	Computation time; number of key nodes; coverage rate	Zhang et al. (2025d)
Complete coverage path planning (CCPP)	Boustrophedon coverage	Improved DRL-based re-DQN	Coverage path length shortened by 31.56%; total navigation time shortened by 35.72%	Path length; total navigation time	Wang et al. (2023b)

(*continued on next page*)

Table A10 (*continued*)

Navigation / planning task	Baseline method	Improved / proposed method	Reported performance improvement	Main improved indicator	Reference
Complete coverage path planning (CCPP)	Conventional coverage planning	Travelling Salesman Problem (TSP) combined with A* search + timed elastic band (TEB)	Overall field completion time shortened by 20%; furrow deviation remained < 10 cm	Completion time; furrow deviation	Ait et al. (2023)
Multi-machine cooperation and task allocation	Independent operation / basic two-vehicle configuration	Multi-machine cooperation with improved ant colony optimisation (ACO)	Path cost of the harvester–transporter system reduced by 27.22%; average operation cycle shortened by up to 67.32%	Path cost; operation cycle	Cao et al. (2021)
Complete coverage path planning (CCPP)	Traditional Boustrophedon or heuristic CCPP	improved genetic algorithm (GA)-based CCPP	Repeated working area reduced by 38.54%; number of U-turns reduced by 35.00%; energy consumption saved by 7.82%	Repeated area; U-turns; energy consumption	Wu et al. (2023)
Path tracking efficiency optimisation	Conventional nonlinear model predictive control (NMPC)	efficiency-oriented model predictive control (EfiMPC)	Operational efficiency improved by 8.56% compared with conventional nonlinear model predictive control (NMPC)	Operational efficiency	Xu et al. (2022a)

Note: The improvement percentages were extracted as reported in the original studies. Direct comparison across rows should be made cautiously because the baseline methods, operating environments, platform scales, and evaluation metrics are heterogeneous.

Appendix B. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.compag.2026.112078.

Data availability

Data will be made available on request.

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